

Re-visiting the Political Economy of the Rise and Fall of the Unidad Popular: Towards a Global Political Economy Approach

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Abstract

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1 Introduction

1.1 In the history of economic thought, progress rarely unfolds as a linear, cumulative ascent toward universal truth. As (Roncaglia, 2006) argues, theoretical advance throughout the history of economic thinking is governed by a *competitive views*: an enduring contest among rival paradigms rooted in fundamentally distinct conceptualizations, analytical categories, and contested world visions. For over half a century, the history of Chile’s Unidad Popular (UP, 1970–1973) under Salvador Allende has been interpreted through an uncritical cumulative lens hegemonized by mainstream economics (Larraín and Meller, 1991; Edwards, 2024). Mainstream accounts under neoclassical market-clearing assumptions and monetarist approach to the dynamics of the balance of payments and inflations precluding the role of distributive conflict, have treated the Chilean breakdown as a cautionary tale of populist fiscal mismanagement. Yet beneath this orthodox consensus lies an unexamined failure of conceptualization: treating peripheral capitalism as a closed monetary system, blind to the structural subordination to the dynamics of the world market in a hierarchically organized world of states featured by and balance-of-payments constraints and financial subordination (Jessop, 2019; Chamorro-Futinico, 2025a).

1.2 This chapter investigates the causal relation between money creation and inflation, and how central bank reserve solvency mediates the inflationary transmission of domestic money creation during a peripheral socialist transformation amidst the global crisis of the north atlantic fordist accumulation regime (Vidal, 2019). If money creation mechanically drove inflation, why did base money expand by over 100% in 1971 while monthly inflation remained quiescent at 1.5–2.0% and annual inflation fell from 35% to 22%? The answer lies in balance-sheet solvency: in 1971, the Central Bank possessed \$400M in inherited foreign reserves and industry operated with ample idle capacity ($\mu \approx 82\%$), absorbing liquidity through real transactions demand. Inflation pass-through did not take place in 1971—it emerged only in mid-1972 once reserves fell below \$30M and the international credit blockade severed revolving commercial trade lines. Broad money growth functioned as an endogenous accommodation valve through which the central bank mediated an balance-of-payments blockade amidst an intense distributive conflict. Contesting the monetarist thesis of Edwards (2024) and the Austrian interpretation of Espinosa and Véjar (2025), I formalize the central bank balance sheet through the Solvency Ratio—the ratio of convertible foreign reserve assets to domestic monetary liabilities ($\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t}$)—identifying it as the state-dependent threshold governing monetary transmission. My formal threshold tests decisively reject linear interpretations of the relation between money creation and inflation [PLACEHOLDER FOR FINAL RESULTS SUMMARY].

1.3 Allende’s thousand days unfolded within the tectonic shifts of the global structural crisis of fordist capitalism during early 1970s. The unilateral suspension of dollar-gold convertibility by the United States on August 15, 1971 dismantled the Bretton Woods system, restoring dollar hegemony (Vernengo, 2021). Simultaneously, the secular post-war decline of the corporate profit rate across the advanced capitalist core (Weisskopf, 1979; Bowles et al., 1983; Vidal, 2019) narrowed the geopolitical room for peripheral developmentalism. The collapse of the Chilean socialist transition represented the structural collision of domestic redistributive aspirations of popular movements as response to the failure of the developmentalist state in its promise of social inclusion at the shadow of the break down of Bretton Woods international monetary order.

1.4 Mainstream and neoclassical syntheses characterize the Allende administration through the paradigm of “*macroeconomic populism*” (Dornbusch and Edwards, 1990; Larraín and Meller, 1991). In this interpretation, redistributive governments inevitably trace an unforced four-phase cycle: initial reactivation through idle capacity, the emergence of bottleneck shortages, unsustainable fiscal and monetary financing, and final economic collapse. By treating the central

bank as an unconstrained printing press and monetary emission as an unmediated political choice, this literature commits two deep epistemological failures. First, it succumbs to what [Blanco and Matute \(2018\)](#) formulate as the *illusion of causality*—a cognitive heuristic that infers direct causal generation from the mere temporal coincidence of salient events (the 1970 election, 1971 wage readjustments, and 1972–1973 inflation) while ignoring non-contingent external covariates. Second, it exercises what [Isch et al. \(2026\)](#) classify as *narrative license*—the rhetorical escalation of observational and correlational snapshots into ironclad causal laws to satisfy academic demands for compelling monocausal parables. By analyzing aggregates in an institutional vacuum, mainstream accounts conflates correlation by causality overlooking the methodological underpinnings of the so-called "credibility revolution and causal identification. Concretely, this conflation manifest itself by misidentifying the symptoms of re-distribution, external blockade, and foremost the exogenous consequences in the balance of payments in the Chilean economy of the unintended unilateral decision by the Federal Reserve of the United States closing of the convertibility window and the collapse of the Bretton Woods international financial regime with the mechanical results of domestic monetary policy.

1.5 To discriminate between these contested claims on rigorous empirical ground, I deploy an applied Lakatosian evaluation of contested scientific research programmes in economic thought ([Lakatos, 1978](#); [King and Sznajder, 2006](#); [Kvangraven, 2021](#)). This methodological framework appraises competing programmes by distinguishing their immutable theoretical axioms (*hard core*) from the auxiliary specifications of their empirical models (*protective belt*), evaluating whether empirical anomalies are absorbed through theoretically progressive extensions that predict *novel facts* or through ad hoc degenerating immunizing stratagems: The debate over the macroeconomic destabilization of the Unidad Popular can be structured into three competing research programmes. The monetarist programme ([Edwards, 2024](#)) anchors its hard core in exogenous base money creation, fractional-reserve multiplier causality, and long-run monetary neutrality in a closed system. Confronted with the empirical anomaly of 100% monetary growth coinciding with non-inflationary output expansion in 1971, its protective belt relies on ad hoc narrative accounts that dismiss the \$294 million external financing cutoff, the U.S. Eximbank embargo, and domestic distribution bottlenecks as secondary disturbances. The Austrian business-cycle programme ([Espinosa and Véjar, 2025](#)) similarly emphasizes domestic distortions, positing that suppressed interest rates and credit expansion generated intertemporal malinvestment and coordination breakdown.

Under the umbrella of dependency theory understood as a *research programme* ([Kvangraven, 2021](#)), rooted in international currency hierarchies, balance-of-payments constraint, endogenous credit money, and exogenous factors in the periphery as expresion of the dynamics of financial subordination ([Chamorro-Futinico, 2025a](#)), I define a new dimensionless index of central bank foreign reserve solvency as an endogenous state variable ($\text{SolvR}_t \equiv e_t \cdot \text{IR}_t / M_t$). This framework allows to identify state dependency of peripheral Central Bank solvency/insolvency in terms of being capable to defend the stability of its own currency. When the central bank maintains a solvent reserve of international currency buffer ($\text{SolvR}_t > 0.21$), domestic liquidity expansion is capable to mobilize idle capacities without triggering inflation, and commercial credit systematically leads primary monetary emission. Progressive pass-through into consumer prices and currency depreciation occurs only when external financial asphyxiation forces the central bank across the critical threshold into balance-sheet insolvency ($\text{SolvR}_t \leq 0.21$), compelling defensive monetary accommodation to sustain enterprise payrolls under severe import starvation.

1.6. This chapter deepens the theoretical foundations established in Chapter 2. Whereas Chapter 2 identified the external boundary of accumulation through an annual balance-of-payments threshold that throttles mechanization under unbalanced growth, this chapter uncovers the dynamic balance-sheet mechanisms through which that external constraint operates at high frequency. To overcome the temporal aggregation bias and closed-economy endogeneity perva-

sive in annual accounts (Dornbusch and Edwards, 1990; Edwards, 2024), my research design assembles two tier data set at annual and monthly frequency time series from multiple secondary sources.

The annual time series data set (1920–2010, $T = 91$) draws functional income shares from Astorga (2023), balance-of-payments accounts from Nieves and Piketty (2025), and disaggregated net capital stocks from the Perpetual Inventory (GPIM) dataset developed in Chapter 2. At this annual frequency, the Austrian malinvestment hypothesis (H_2) fails decisively: gross fixed investment in machinery and equipment contracted (-2.3%) during the 1971 expansion while structural capacity utilization surged to an all-time peak of $\mu_{1971} = 98.44\%$, presenting empirical evidence favorable to the hypothesis of reactivation rather than intertemporal capital misallocation.

To isolate the transmission belts of the subsequent breakdown, the high-frequency monthly tier (1960–1980, $N = 250$ continuous months) integrates primary Central Bank of Chile archival data accessed through the Application Programming Interface (API) of the *Base de Datos Estadísticos* (BDE) of the Chilean Central Bank (BCH) including Central Bank base monetary emission H and $M1$ including deposits and less liquid measures of money, indices of gold, copper and wholesale US prices, Indexes of export, import, manufacturing, and mining activity series here gathered by (Díaz-Bahamonde, 2023) and included in the data set to built import capacities and structural imbalances indices. IMF Central Bank reserve assets historical statistics provided stocks of international reserves at monthly frequency that cannot find elsewhere. Following the August 1971 close of FED gold convertibility window, my proxy of Prebisch’s purchasing power capacity to import (M_t^{cap}) fell to 27.7, driving the Real Structural Imbalance Ratio ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$) to an unsustainable peak of 283.4 in May 1973. In our Toda–Yamamoto and Threshold VAR systems, this physical import strangulation governs the regime switch: once foreign credit embargoes deplete foreign exchange reserves below the identified critical solvency threshold ($\text{SolvR}_t \leq 0.21$), external price shocks penetrate domestic retail prices immediately, compelling defensive monetary accommodation to maintain enterprise payrolls amidst supply paralysis.

1.7 REVISE ONCE FINISHED OTHER SECTIONS The remainder of this chapter is structured as follows. Section 2 surveys the contested traditions in Chilean political economy and formalizes the epistemological critique of orthodox causality. Section 3 develops the theoretical framework of World Money, monetary endogeneity, and the open-economy transposition of Minskyan solvency regimes to the peripheral central bank. Section 5 grounds the inquiry in the historical political economy of the pre-1970 accumulation regime, details the multi-frequency data architecture, specifies the model restrictions, and implements the econometric evaluation across the four-stage empirical strategy. Section 6 discusses the political economy of peripheral solvency, situates the UP within the global structural crisis of the 1970s, and outlines lessons for contemporary monetary sovereignty.

2 Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular

2.1 No single framework commands consensus on the political economy of the rise and fall of Chile’s Unidad Popular (UP). Instead, rival explanatory traditions have structured the field, each emerging under unequal political conditions and distinct historical contexts. Transnational networks linking the Pontificia Universidad Católica de Chile (PUC) to the University of Chicago formed the neoliberal school, which subsequently acquired state power through the dictatorship that followed the 1973 coup. Neo-structuralism emerged under the dictatorship as

a technocratic opposition to Pinochet’s regime—specifically in the aftermath of the 1982 debt crisis—before rising to government power and establishing itself as the mainstream economic paradigm during the post-dictatorship order. Contemporary accounts in the field of Historical Political Economy (HPE) entered the debate much later, bringing powerful tools of causal identification to historical questions. However, they did so within a disciplinary environment already shaped by earlier causal claims taken as given, while perspectives anchored in distributive conflict and structural dependency were seldom considered or actively excluded. Dependency theory occupies a different position altogether. It was the intellectual landscape from which the UP programme itself emerged, and the coup institutionally shattered it through violent repression before it could resolve the theoretical problems it had opened up.

2.2 These traditions did not simply coexist. They succeeded one another through political conflict without genuinely opposing each other on the intellectual terrain, despite stark declarations regarding the intellectual bankruptcy of the structuralist-dependency school:

“...the macroeconomic view of the Unidad Popular failed to recognize that their policies would only be able to generate a burst of economic activity that would be unsustainable in the medium term... Needless to say, this view of the way the economy functioned ignored many of the key principles of traditional economic theory.” (Dornbusch and Edwards, 1989, p. 16–17).

Each identified a part of the problem, yet none integrated external transmission, distributive conflict, and macroeconomic dynamics into a single explanatory sequence. This section critically evaluates these traditions. It reconstructs the historical conditions under which each approach became authoritative, clarifies the mechanisms each privileged, and shows which questions each rendered visible and which it displaced, sanitized, or foreclosed. It begins with classical dependency theory, which posed the problem the chapter as a whole seeks to resolve more sharply than any approach that followed.

2.3 Consequently, this fragmented intellectual legacy has resulted in a striking scarcity of rigorous empirical evidence concerning the macroeconomic dynamics of the UP’s collapse. While the political and institutional histories of the period are extensively documented, the macroeconomic debate has largely relied on retrospective narratives, analysis of correlation without causality, or ideological posturing rather than methodologically robust quantitative analysis. There is a critical lack of empirical studies employing adequate econometric methods to isolate and measure how allegedly “*populist interventions*”—such as exchange rate overvaluation, fiscal monetization, and generalized price controls—transmitted distributive conflicts and external shocks into systemic macroeconomic failure. By leaving these macroeconomic transmission mechanisms empirically untested, the literature has remained trapped in theoretical stalemates. This chapter seeks to bridge this methodological void by moving beyond ideological retrospection, advancing more rigorous empirical identification to the macroeconomic data of the UP era to trace the anatomy of its economic unraveling.

2.1 Classical Dependency Theory: Interrupted Programme

2.3 Dependency theory did not approach the Unidad Popular from the outside. It helped constitute the intellectual horizon within which the UP became thinkable. Its formation in Chile brought together two lineages that converged, institutionally and theoretically, in Santiago during the long sixties. From one side came the Monthly Review tradition, beginning with Baran (1957)’s analysis of economic surplus under monopoly capitalism and radicalized by Frank (1967) into the development-of-underdevelopment thesis: underdevelopment was not a residual stage but an active product of incorporation into the world capitalist system. From the

other came the structuralist school built within the headquarters of the Economic Commission for Latin America (ECLAC), led by Raúl Prebisch, which had already framed Latin American development through centre–periphery relations, external bottlenecks, and the historical asymmetries of international trade.

2.4 At the Centro de Estudios Socioeconómicos (CESO) of the Universidad de Chile, these two lineages fused into an active research programme. Following the 1964 military coup in Brazil, the arrival of Theotônio dos Santos and Vania Bambirra as political exiles in Santiago catalyzed this institutional convergence (Cárdenas et al., 2022). Collaborating at CESO alongside André Gunder Frank, Orlando Caputo, and Roberto Pizarro—and maintaining close dialogue with Ruy Mauro Marini in Concepción—they transformed dependency from a general critique of peripheral capitalism into an effort to specify the historical forms linking regional subordination, capital accumulation, and political strategy (Dos Santos, 1970; Bambirra, 1971; Caputo and Pizarro, 1970; Kay, 2010). Far from remaining an academic abstraction, this intellectual collective directly shaped the policy architecture of the Chilean left: among Dos Santos and Bambirra’s closest collaborators and disciples were Sergio Ramos and Alexis Guardia, the primary economic drafters of the Unidad Popular programme representing the Communist and Socialist parties, respectively.

2.5 The programme’s analytical wager was clear. Peripheral capitalism followed a distinct developmental dynamic structured by external constraints, a domestic bourgeoisie structurally incapable of autonomous national accumulation, and reproduction patterns that continually re-entrenched subordination. Within this framework, socialist transformation appeared as an objective developmental necessity grounded in the physical limits of peripheral capital formation. Yet the tradition never achieved theoretical unity. Marini’s theory of superexploitation pushed the analysis toward a world-market logic where the systematic compression of labor below the value of its reproduction conditioned domestic capital accumulation (Marini, 1973). Conversely, Cardoso and Faletto (1979) insisted that dependency took historically contingent forms governed by domestic class alliances and the relative autonomy of the capitalist state. This tension proved decisive. At stake was whether domestic political mediation constituted an independent object of analysis or simply mediated world-market imperatives. Classical dependency posed the question with great force but never resolved it.

2.6 That unresolved tension did not prevent the tradition from producing the richest contemporaneous political-economy accounts of on the Political Economy of the Rise and Fall of the UP. De Vylder (1976)’s *Allende’s Chile* provided the seminal participant-observer account, correctly locating the material foundation of the external constraint in Chile’s extreme physical import dependency: over 90% of capital equipment was imported, and more than 95% of machinery in large-scale copper mining originated in the United States. Yet De Vylder’s framework suffered from a fatal analytical concession. Confronted with orthodox arguments attributing the crisis entirely to domestic mismanagement, De Vylder retreated into an internal-blame explanation, asserting that:

“the problem was the insatiable imports, not the US blockade nor the lack of credit”
(De Vylder, 1976, p. 106).

To sustain this concession, De Vylder pointed to gross nominal credit lines extended by the Soviet Union and Eastern Europe (> \$400 million) as ostensible proof that external financing remained abundant, conflating gross nominal commitments with usable, convertible liquidity.

2.7 Stallings (1978) overturned De Vylder’s concession by uncovering the *qualitative non-fungibility of international credit*. Drawing on primary Central Bank foreign exchange ledgers, she demonstrated that bilateral socialist credits were non-convertible tied lines earmarked for specific industrial equipment. They could neither purchase patented American spare parts for

Kennebecott copper haul trucks nor settle immediate spot cash contracts for wheat and petroleum on Western commodity markets. Concurrently, private U.S. commercial banks slashed revolving short-term trade credit lines from \$220 million to \$35 million, while the Export-Import Bank froze disbursements. These debates around balance-of-payments constraints did not unfold in isolation. They formed part of a vibrant contemporaneous intellectual landscape captivated by the *vía chilena al socialismo*—the unprecedented historical attempt to achieve revolutionary socialist transformation through bourgeois democratic institutions. Across this field emerged a rich empirical corpus tracking the crisis from complementary angles: distributional conflict across successive administrations, worker and peasant self-organization from below, the logistical spare-parts embargo, the monetary and balance-of-payments mechanisms through which external pressure became domestic crisis, and the strategic dilemmas of the state under conditions of emerging dual power (Stallings and Zimbalist, 1975; Espinosa and Zimbalist, 1978; Kay, 1978; Petras and Morley, 1975; Griffith-Jones, 1981; Sideri, 1979; Sweezy and Magdoff, 1974; Zavaleta Mercado, 1974; Winn, 1986). The enduring methodological strength of classical dependency lay precisely here: it refused to treat external imperial intervention and domestic class struggle as competing explanations, approaching them instead as interacting, co-constitutive dimensions of the same historical process.

2.8 That achievement also marked the analytical limit dependency theory could not overcome in its classical form. Bounded by a bilateral nation-state framework (the U.S. versus Chile), it omitted the international monetary architecture of Bretton Woods and global stagflation, while lacking a formal macroeconomic specification linking external balance-sheet shocks to domestic distributive conflict and capacity utilization. Yet the defeat of dependency theory was twofold. Politically, the coup dismantled the institutions, networks, and research centres in which the programme had been elaborated. Intellectually, the interruption arrived before the tradition could work through its own internal tensions. No superior explanatory framework refuted it in a scientific terrain. It broke under conditions in which theoretical incompleteness coincided with the collapse of Chile’s liberal democracy, the failure of democratic socialism to resolve the crisis under organized domestic and international sabotage, and the installation of a subsequent civic-military dictatorship. Subsequent approaches entered a field already reorganized by force, not reason. The next school to become authoritative in Chilean political economy was not the one that best resolved the problem dependency had identified, but the one that arrived coordinated with the counterrevolutionary historical bloc that constituted the reconstruction of the state and civil society Casals (2021).

2.2 The Rise of the Chicago Boys and the Neo-liberal School

2.9 The neoliberal school did not emerge from within the intellectual world of the Unidad Popular as one competing interpretation among others. It entered through a different route: as a transnationally trained programme of economic reconstruction that acquired authority through the counterrevolutionary reorganization of the Chilean state. Its genealogy is well established. The 1956 agreement between the Pontificia Universidad Católica de Chile and the University of Chicago opened the institutional channel through which Milton Friedman, Arnold Harberger, and Gary Becker trained a cohort of Chilean economists in monetarism, public finance orthodoxy, and neoclassical price theory. By the late 1960s, this network had evolved from an academic outpost into a politically embedded project, closely aligned with business associations and the Chilean right, oriented toward the systemic restructuring of Chilean capitalism (Valdés, 1995; Fischer, 2009; Clark, 2017).

2.10 That programme found its canonical expression in *El Ladrillo*². The text codified in advance the macroeconomic blueprint on which the post-UP order would be constructed. It

²*El Ladrillo* (“The Brick”) was drafted clandestinely prior to the September 1973 coup by Chicago-trained

operated as a counterrevolutionary vanguard design: an operational policy manual prepared prior to the coup to direct the reconstruction of Chilean capitalism the moment the popular government was overthrown.

2.11 Its diagnosis identified a familiar catalogue of structural problems: chronic inflation, recurrent external deficits, agricultural stagnation, low productivity growth, and distortions associated with state-led industrialization. Several of these observations captured real tensions within Chilean accumulation, and they merit serious engagement. The argument that prolonged relative-price controls hindered export diversification and exacerbated balance-of-payments vulnerability cannot simply be dismissed, nor can the recognition that import-substitution industrialization had reached severe structural exhaustion (De Castro, 1992). Yet the broader theoretical hypothesis holding these observations together was much weaker: the claim that the crisis of the UP merely culminated a secular drift toward the excessive socialization of the economy. That thesis recoded a complex conjunctural crisis into a normative indictment, naturalizing market discipline rather than explaining the empirical dynamics of the breakdown.

2.12 Here lay the neoliberal school's constitutive limit. It registered macroeconomic imbalances only by systematically stripping them of their political-economy. Functional income distribution appeared not as structural class conflict, but as a technical distortion of price signals. The external crisis was treated as the endogenous consequence of fiscal and monetary mismanagement, never as the relational interaction between domestic class struggle and international financial subordination and destabilization. The October 1972 employers' lockout, the private commercial credit blockade, the spare-parts embargo, and covert foreign destabilization could not enter the account without undermining the framework's own core hypothesis and axioms. Neoliberalism did not omit these factors by accident. Its explanatory logic required them to disappear, since acknowledging them would have undermined the premises that the crisis proved the inherent impossibility of socialist transformation.

2.13 The neoliberal interpretation became authoritative under historical conditions that were not primarily intellectual. Its ascendancy reflected no superior theoretical integration of external transmission, distributive conflict, and macroeconomic dynamics. A programme formulated before the coup entered a terrain reorganized by military force, and the dictatorship installed it as the common sense of national economic reconstruction. That restructuring violently dismantled collective bargaining, political mediation, and labor organizations while relying, paradoxically, on the centralized administrative apparatus of the developmental state it claimed to dismantle (Clark, 2017; Bockman, 2019). The neoliberal school was less a scientific solution to the developmental dilemmas of the periphery than a victorious political project presenting ruling class backlash as economic science. Its enduring legacy lies precisely here: it converted a conjunctural crisis resolved by state terror into a universalized economic doctrine asserting the impossibility of any non-market alternative (Espinosa, 2021).

2.3 The Neo-Structuralist School: Selective Sanitization

2.14 The neo-structuralist school was an intellectual accomplishment of the post-dictatorship order rather than a simple continuation of classical structuralism. It established its institutional base during the dictatorship, primarily within the CIEPLAN think tank, where former structuralists and democratic opposition economists critically monitored the Chicago Boys' reforms while formulating an alternative programme for the democratic transition. That genealogy explains much of the school's subsequent authority. The CIEPLAN economists were more than

economists coordinated by Emilio Sanfuentes, with contributions from Sergio de Castro and Pablo Baraona, in liaison with naval officers and the leadership of the private employers' association *Sociedad de Fomento Fabril* (SOFOFA) (De Castro, 1992).

political moderates who had tempered their commitments under authoritarian rule (Silva, 1991). They constituted the modernizing technocratic fraction through which the post-authoritarian centre-left could challenge orthodox monetarism while operating strictly within the boundaries of market-disciplined, export-oriented peripheral capitalism. Neo-structuralism entered the field as a politically functional renovation calibrated to manage Chile's subordinate insertion into the globalized economy (Sklair, 2000; Leiva, 2008; Maillet et al., 2016).

2.15 The decisive theoretical operation of that renovation was the abandonment of the classical surplus approach. Where Prebisch and the foundational ECLAC tradition had centered the production, appropriation, and reinvestment of the economic surplus under peripheral conditions, neo-structuralism recast underdevelopment into the mainstream idiom of market imperfections, institutional bottlenecks, and pragmatic state regulation. This shift did not merely modernize structuralism; it altered its analytical object. The external constraint remained, but strictly as a macroeconomic financing ceiling rather than as a structural relation of surplus extraction and dependent reproduction. By reducing a theory of historically structured subordination to a technical problem of policy calibration, the school depoliticized the external barrier. As Leiva (2008) demonstrates, without the surplus approach, the external constraint survives only in a thinned accounting form: legible as a foreign exchange deficit, but stripped of the class relations that generate and sustain it. Neo-structuralism did not forfeit empirical rigor; it forfeited the conceptual framework required to theorize the balance of payments and distributive conflict as unified moments of the same accumulation process.

2.16 The school's account of the UP crystallized most authoritatively in the work of Sergio Bitar. Originally published in 1979, Bitar (2013) provided the most sophisticated neo-structuralist reconstruction of the popular government. Its principal merit lay in rejecting an exclusively monetary explanation of the external accounts. Bitar sought to integrate foreign exchange scarcity, political opposition, and institutional transformation into a single dynamic narrative. His formulation of a "viability area" remains a genuine analytical contribution, demarcating the narrow economic margin within which the UP programme could have advanced without precipitating macroeconomic collapse. Yet that contribution also established the school's defining silence. Across multiple editions, Bitar never substantively engaged the contemporaneous dependency literature that had developed across the same historical terrain. Foundational accounts by De Vylder (1976) and Stallings (1978) were already widely available, yet Bitar avoided confronting them. This silence was neither a bibliographic omission nor a conspiracy. It reflected the strategic selectivities accompanying the abandonment of the surplus approach (Leiva, 2008), marking the boundary at which dependency's central questions were cordoned off from post-dictatorship economic discourse.

2.17 That theoretical cordon became hegemonic through the populist-macroeconomic paradigm. Larraín and Meller (1991) codified the UP into a stylized four-phase trajectory of initial reactivation, emerging bottlenecks, severe destabilization, and ultimate collapse. Dornbusch and Edwards (1991) then generalized this sequence into a regional archetype of "macroeconomic populism." This paradigm achieved wide influence by providing a clear narrative, identifiable turning points, and cautionary lessons that circulated throughout Latin America during its democratic transitions. Yet it operated through a definitional foreclosure: by classifying redistributive policies categorically as "*macroeconomic populism*," the framework ruled out the possibility that wage-led growth could ever be structurally sustainable under peripheral conditions. Meller (2000) subsequently cemented this interpretation as the common sense of the UP: an unforced tragedy driven by policy mismanagement and political polarization. The result was a reading self-critical in tone but exculpatory in effect, reducing the crisis to domestic excess while treating external financial blockades and domestic logistical sabotage as secondary noise. The contrast between Sergio Ramos (1979) and Joseph Ramos (1980) illustrates this epistemological divide: the former treated inflation as a structural contradiction internal to a socialist

transition under acute class struggle, whereas the latter reduced the crisis to an engineering problem of disequilibrium management and stabilization failure.

2.18 Neo-structuralism thus substituted a technocratic problem of governance for the conflict-centered political economy generated during the UP itself (Larraín and Meller, 1991; Dornbusch and Edwards, 1991; Meller, 2000). The school’s analytical limits were not accidental blindspots; they were strategic exclusions built into the conditions of its post-authoritarian legitimacy. Its primary historical function was to preserve the structuralist language of external constraints while adapting it to the macroeconomic governance of a neoliberalized economy, sustaining two decades of centre-left political stability. In doing so, it transformed the thousand days of Allende’s government into a moral tale of redistributive overreach. Neo-structuralism became the progressive framework through which the post-dictatorship order could be managed and stabilized without reviving the class-conflict categories that made earlier structuralist and dependency traditions dangerous to ruling-class hegemony (Leiva, 2008).

2.19 This critique does not dismiss the empirical contributions of scholars such as Bitar or Meller; it specifies the political-economic renounces and strategic selectivities as consequence of their synthesis. The coordination of monetary, structural, and political variables into a coherent narrative remain important contributions to the debate. However, the systematic exclusion of class struggle for the economic surplus was the price of its institutionalization during the democratic transition and the post-dictatorship period. That compromise defines neo-structuralism’s enduring legacy. It established the default macroeconomic memory inherited by subsequent scholarship, leaving historical political economy equipped with sophisticated empirical tools, but lacking a coherent theory of the structural mechanisms at work.

2.4 Historical Political Economy: Methods Under Inherited Macro Theory

2.20 Historical Political Economy (HPE) represents a genuine methodological advance in the study of economic history. By applying the credibility revolution’s identification toolkit to historical inquiry, it has introduced a level of empirical precision rarely achieved in earlier debates. Its institutionalization around the *Journal of Historical Political Economy* and the *Oxford Handbook of Historical Political Economy* reflects a broader effort within economics to reclaim historical questions without sacrificing rigorous causal identification (Jenkins and Rubin, 2024; Charnysh et al., 2023). Applied to twentieth-century Chile, this agenda has generated an impressive corpus examining price controls, real wages, land redistribution, and political violence.

2.21 Yet HPE did not generate the archival terrain on which it operates. That foundation was constructed by a parallel renewal in Chilean economic history, which has reconstructed long-run time series on prices, trade, living standards, and distribution. The archival work of Mario Matús, Manuel Llorca-Jaña, Claudio Robles, and their collaborators established the quantitative records that made micro-econometric identification feasible. Within this archival foundation, Rodríguez Weber (2018) is foundational for the empirical architecture of this dissertation: the long-run historical accounts of inequality compiled there directly underpin the functional wage-share series (ω_t) harmonized by Astorga (2023) and utilized throughout Chapters 2 and 3. By situating the Unidad Popular within a secular century-long cycle of distribution and explicitly demonstrating how political coercion shaped income shares, Rodríguez Weber provides a crucial historical bridge between institutional economic history and structural macroeconomic questions.

2.22 In contrast to descriptive economic history, HPE proper focuses on estimating discrete causal effects. Aldunate et al. (2022) exploit firm-level records to identify the credit-allocation shocks induced by U.S. financial warfare, while Jaimovich and Toledo-Concha (2021) trace

contemporary Mapuche territorial conflict to the localized geography of agrarian expropriation and military counter-reform. [Girardi and Bowles \(2018\)](#) provide a distinctive contribution: their high-frequency event study of the Santiago stock exchange demonstrates that financial wealth-holders responded systematically to shifts in the political threat to private property, corroborating classical political economy’s focus on capitalist investment strikes. These studies represent sophisticated empirical contributions. However, they perform fundamentally different analytical work: some build archival ledgers, others isolate localized causal effects, while few—if not [Girardi and Bowles \(2018\)](#) alone—examine macroeconomics and institutions through a theoretical framework centered on the structural power of capital over the economy.

2.23 This analytical boundary is most evident in the work of Felipe González and his collaborators, who define the contemporary frontier of Chilean HPE ([González, 2013](#); [González and Vial, 2021](#); [Aldunate et al., 2022](#); [González et al., 2020](#); [Bautista et al., 2023](#); [Cuesta et al., 2015](#); [González and Prem, 2026](#)). Their research is methodologically exemplary. They have unearthed primary archival records, formulated transparent quasi-experimental designs, and resolved contested empirical questions surrounding land reform, industrial requisitions, covert foreign intervention, and the institutional legacies of the military dictatorship. Yet this corpus also demarcates the field’s constitutive limit: the macroeconomy appears primarily as a passive background upon which institutional interventions and policy shocks are measured, rather than as an organic, contradictory system of capital accumulation, surplus extraction, and crisis. Even where the archival evidence touches systemic macroeconomic phenomena, the theoretical interpretation is routinely imported from post-dictatorship mainstream narratives. The result is scholarship of exceptional empirical precision that displaces political economy in the classical sense into an applied microeconomics of institutional interventions.

2.24 This boundary reflects the field’s disciplinary formation. HPE arrived at the Chilean case after the historical clash between the dependency, neo-structuralist, and neoliberal paradigms had already been settled by state coercion. Consequently, it uncritically inherited the depoliticized macroeconomic vocabulary of the post-dictatorship consensus, which reduced the UP’s crisis to a cautionary tale of macroeconomic populism and unforced balance-of-payments mismanagement. The research programme developed in this chapter builds upon HPE’s archival and empirical rigor, while restoring what applied microeconomics omits: an integrated theoretical framework capable of modeling external balance-of-payments vulnerability, distributive conflict, and domestic monetary accommodation as mutually conditioning moments of peripheral capital accumulation.

2.5 The Revival of Dependency Theory as a Research Programme

2.25 The return to dependency theory is not an exercise in intellectual nostalgia. It recovers a research programme whose historical retreat signaled a profound collapse of the region’s sociological imagination. As [Palma \(2016\)](#) argues, Latin America’s critical thinking swung like a pendulum from one type of absolute certainty to another: from the mechanical stagnationism of early dependency to the uncritical market fatalism of the post-dictatorship consensus. In that shift, the region’s intellectual landscape was stripped of its capacity to conceive development as a problem of collective agency, structural transformation, and sovereign autonomy. Development was degraded from a political struggle over the production, appropriation, and reinvestment of the economic surplus into an administrative exercise in macroeconomic compliance. The analytical task today is neither to revive the rigid certainties of the 1970s nor to surrender to technocratic fatalism, but to reconstruct dependency as an empirically generative, progressive framework.

2.26 [Kvangraven \(2021\)](#) provides the epistemological architecture for this reconstruction by

formalizing dependency theory through Imre Lakatos’s methodology of scientific research programmes. Kvangraven refutes the mainstream caricature of dependency as a monolithic dogma by demonstrating that the tradition is structured around a coherent, shared *hard core*: the premise that global capitalist accumulation systematically reproduces hierarchical structures between core and periphery that constrain the developmental space of subordinate social formations. The scientific vitality of the programme does not depend on dogmatic consensus regarding specific historical models; rather, it hinges on whether its *protective belt of auxiliary hypotheses* can generate *progressive problemshifts*—predicting novel empirical facts and specifying concrete causal mechanisms—rather than degenerating into defensive ad-hoc immunizations. By formalizing this distinction, she clarifies how dependency can assimilate new historical evidence within the context of the concrete realities of global capitalism without abandoning its core ontological commitments.

2.27 Madariaga and Palestini (2021) sharpen this epistemological reconstruction by distinguishing between *situations of dependency* and *mechanisms of dependency*: i.e., between historically specific descriptive configurations and the causal processes that reproduce external constraints. This distinction has energized a contemporary renewal in Santiago, where scholars have operationalized dependency theory to analyze rentier accumulation, export stagnation, and dependent financialization in present-day Chile (Ahumada, 2019; Ruiz Bruzzone, 2021; Ahumada and Torres, 2024; Sossdorf Gonzalez, 2024). Within this revival, recovering the historical memory of the Unidad Popular is not an exercise in political nostalgia, but an inquiry into the structural conditions under which sovereign control over the surplus and democratic self-determination were attempted and violently suppressed (Arboleda, 2023; Ahumada and Chang, 2025). Yet for the macroeconomic analysis of the UP itself, the crucial task remains unfinished: while classical accounts identified the concrete *situation* of Chilean vulnerability (import dependency and bilateral credit retaliation), they lacked a formal *mechanism* linking peripheral balance-of-payments asphyxiation to the structural crisis of the core’s international monetary order.

2.27 This missing transmission mechanism can be formulated by synthesizing the political economy of global dollar hegemony with the institutional mechanics of the dissolution of Bretton Woods. Vernengo (2021) provides the structural foundation for why Bretton Woods collapsed: a macro-political drive by U.S. elites to escape the discipline of gold convertibility, discipline domestic labor through international capital mobility, and establish an unconstrained fiat dollar standard. In turn, Zoeller (2019) demonstrates how and when this rupture unfolded: rather than a predetermined grand design, closing the Federal Reserve convertibility gold window was a contingent institutional maneuver forced by IMF institutional veto points and European surplus resistance. The core executed an emergency exit on August 15, 1971, rapidly consolidating a floating dollar regime that eliminated its own reserve discipline while causing an exogenous shock to peripheral economies.

2.28 For Chile, this historical timing was decisive. Nixon’s unilateral closure of the gold window occurred exactly three days after the U.S. Export-Import Bank formally denied credit lines to the Unidad Popular on August 12, 1971. Peripheral accumulation and the “*via Chilena al Socialismo*” was thus caught in converging external constraints: the bilateral credit non-fungibility documented by Stallings (1978) colliding with the systemic dissolution of the global monetary regime. Incorporating this dual transmission into the protective belt of dependency theory resolves the theoretical ambiguities that stalled its classical formulation. It equips the tradition with an exogenous event (Nixon’s administration closing the Fed convertibility gold window), as a macroeconomic mechanism capable of explaining how external balance-sheet shocks triggered domestic supply paralysis and forced monetary accommodation, directly setting up the hypothesis tournament among contested approaches to economic thinking to be developed in this chapter.

2.6 Epistemological Critique: The Illusion of Causality, Narrative License, and Collider Traps

2.33 The persistent divergence among these explanatory traditions cannot be resolved through historiographical narrative alone. As [Blanco and Matute \(2018\)](#) establish, human causal reasoning is systematically vulnerable to the *illusion of causality*: a robust cognitive bias wherein observers detect strong causal links between co-occurring events even when the statistical contingency is zero or driven by unobserved common causes. In the economic historiography of the Unidad Popular, orthodox monetarist and Austrian accounts observe a conspicuous historical coincidence—the election of a socialist government in late 1970, expansionary fiscal-wage policy in 1971, and the explosion of inflation in 1972–1973. Relying on intuitive heuristics, they assign exclusive causal responsibility to domestic monetary issuance ($g_{M,t} \rightarrow \pi_t$), treating international monetary hierarchy, terms-of-trade shifts, and geopolitical destabilization as irrelevant background noise ([Edwards, 2024](#); [Espinosa and Véjar, 2025](#)).

2.34 This causal illusion is reinforced by what [Isch et al. \(2026\)](#) identify as *narrative license* and *overreaching causal claims* in historical social science. Orthodox accounts dress up doctrinal priors as ironclad empirical laws without testing alternative causal mechanisms nor specifying counterfactual identification paths. For example, [Espinosa and Véjar \(2025\)](#) assert an Austrian Business Cycle mechanism wherein artificial credit expansion generated an unsustainable capital malinvestment boom. Yet they advance this claim without presenting positive evidence of gross fixed capital deepening (ΔK), ignoring that Chilean fixed machinery investment contracted during 1971 while capacity utilization reached historical peaks. Narrative license allows the theorist to impute unobserved microeconomic mechanisms (capital lengthening) to explain macroeconomic outcomes while dismissing external interventions as quantitatively negligible.

2.35 Methodologically, orthodox accounts fall into what causal inference defines as a *collider trap* ([Pearl, 2009](#)). A collider trap takes place when analysts focus on a symptom that is actually the shared outcome of two separate causes. By isolating their attention on this shared outcome, they distort the relationship between the causes, often falsely attributing the symptom entirely to just one of them. In narratives of the UP crisis, this shared outcome—the collider—is the widespread presence of retail shortages and food queues (C) ([Edwards, 2024](#)).

Orthodox accounts treat these empty shelves as definitive proof of “repressed inflation,” attributing them solely to the government printing money to stimulate demand (U_1). However, these shortages were simultaneously driven by a second, independent factor: the systematic hoarding and lockouts organized by private wholesale distributors and trucking strikes (U_2). Both state demand stimulus (U_1) and private supply lockouts (U_2) collided to produce empty retail shelves (C):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp U_2 \mid C \quad (1)$$

By treating retail shortages as a neutral, objective measure of excess liquidity (conditioning on C), orthodox accounts commit a collider error of causal identification. This methodological blind spot produces a spurious conclusion, allowing them to blame public deficit monetization for a supply paralysis that was heavily driven by the political actions of private business federations.

2.36 To transcend these causal illusions and collider traps, I adapt the methodological approach of [King and Sznajder \(2006\)](#), who formally tested competing economic theories to explain Poland’s growth trajectory during neoliberalization. In this chapter, I treat Monetarism (H_1), the Austrian Business Cycle (H_2), and Dependency Theory (H_3) as rival scientific research programmes. Each possesses a distinct theoretical core, auxiliary assumptions, and falsifiable empirical predictions. I submit the central causal claims of these competing traditions to a formal *Hypothesis Tournament* evaluated across multi-frequency time series data.

3 World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet

3.1 World Money, the Universal Equivalent, and the Monetary Expression of Value

3.1 In Marxian political economy, money originates as the commodity form of the universal equivalent: the unique social medium whose function is to express the value of all commodities and serve as the general means of payment (Basu, 2022, §3.1). Gold historically occupied this role not through inherent natural properties, but through a social process of market validation, acting as the bearer of abstract human labor.

3.2 The historical transit to credit money transforms this institutional form without dissolving its underlying function. As banking systems developed to intermediate commercial credit, central bank liabilities emerged as the legal anchor of domestic circulation (Basu, 2022, §3.2.3). The commodity anchor disappeared from domestic circulation, but the imperative of social validation remained. Central bank liabilities and bank deposits perform the role that bullion once fulfilled: providing the monetary expression through which private, decentralized labor is validated as social labor. Following the Monetary Expression of Value (MEV) framework (Foley, 1982; Basu, 2022), the value of money links aggregate labor-time to the circulating monetary unit within a national formation.

3.3 Once circulation crosses national borders, however, credit money confronts the world market. While Marx presented world money in the physical form of bullion shedding national garbs (Ivanova, 2013, p. 51), a global credit-money system reproduces the universal equivalent through an international currency hierarchy (Vasudevan, 2009; Ivanova, 2020). National currencies are stratified by their degree of international acceptability, liquidity, and convertibility. World money is the historically specific form wherein the liabilities of the dominant hegemonic power operate as the universal means of international payment, reserve asset, and unit of account.

3.2 The Bretton Woods Architecture and Peripheral Balance-of-Payments Constraints

3.4 The post-war Bretton Woods architecture codified this currency hierarchy institutionally. By pegging the U.S. dollar to gold at \$35 per ounce while anchoring other currencies to the dollar, the international monetary order formalized the dollar as the operative form of world money (Vasudevan, 2009, p. 483). This arrangement granted the hegemonic center an *“exorbitant privilege”*: the ability to borrow internationally in its own currency, finance external deficits without reserve constraints, and act as banker to the world (Vasudevan, 2019, pp. 243–244).

3.5 For peripheral economies, this institutional architecture generated an acute structural asymmetry. Peripheral states cannot settle international accounts in their own domestic currencies; external liabilities and essential imports must be discharged in world money. The peripheral circuit of capital confronts an absolute balance-of-payments constraint (Vasudevan, 2016; Abeles, 2013). Domestic credit expansion is bounded by the capacity to command foreign exchange through export earnings, terms of trade, and access to international credit lines. Reserve accumulation constitutes an externally enforced structural prerequisite of economic reproduction (Vasudevan, 2019, p. 244).

3.6 The unilateral closure of the gold window by the United States on August 15, 1971 shattered the fixed-parity anchor of Bretton Woods agreements leading to their demise. While this

rupture removed gold convertibility, it preserved dollar centrality, ushering in an era of global floating fiat money (Vasudevan, 2009, p. 486). For the nascent socialist project of the Unidad Popular, this systemic shock dismantled the international monetary stability upon which its initial macroeconomic projections rested.

3.3 The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Balance-Sheet Solvency

3.7 Financial subordination materializes concretely on the balance sheet of the peripheral central bank. Mainstream open-economy models interpret this balance sheet through the lens of “*reserve adequacy*” and precautionary self-insurance, treating international reserves as an inventory buffer optimized against friction costs to insulate the economy from sudden stops (Jeanne, 2007; Céspedes and Chang, 2024). This normative framing assumes that reserve accumulation represents the voluntary, unconstrained choice of a sovereign optimizing agent.

3.8 In peripheral capitalism, reserve trajectories are the structural outcome of the balance-of-payments constraint, unequal exchange, and financial subordination (Vasudevan, 2009; Abeles, 2013). The two sides of the peripheral central bank’s balance sheet reflect fundamentally distinct monetary forms. On the liability side, domestic money represents unanchored credit obligations that the financial system owes the public, underwritten by state authority (Basu, 2022, §3.2.3). On the asset side, international reserves (IR_t , measured in USD millions) consist of claims on world money that the peripheral state cannot emit, but Central Bank holds as assets against a liability of the United States federal reserve with its own population. In other words, money is credit of the Central Banks with their own populations, but world money is an asset when hold by Central Banks abroad in form of international reserves. .

3.9 This balance-sheet structure is governed by the Central Bank Solvency Ratio:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} \quad (2)$$

where e_t is the nominal exchange rate (Escudos/Pesos per USD), IR_t denotes international reserves in USD, and M_t represents domestic monetary liabilities.³ The dynamic law of motion governing this ratio is:

$$\Delta \text{SolvR}_t = \frac{CA_t + KA_t}{M_t} - \text{SolvR}_t \cdot g_{M,t} \quad (3)$$

where CA_t is current account net receipts, KA_t is net capital flows, and $g_{M,t}$ is domestic money growth. Financial subordination acquires its operative expression through this solvency relation: movements in SolvR_t govern the corridor within which domestic macroeconomic policy can maneuver. The complete formal accounting identities governing this balance-sheet relation are documented in Appendix A.

3.4 Money Endogeneity vs. Exogeneity and the Peripheral Paradox

3.10 Grounding the peripheral balance sheet requires resolving the foundational debate over the causality between bank credit, deposits, and central bank reserves (Rochon and Rossi, 2013; Lavoie, 2014). Mainstream monetarism and orthodox macroeconomics operate under

³The Solvency Ratio is the inverse of the Leverage Ratio ($\text{LevR}_t \equiv M_t/(e_t \cdot IR_t) = \text{SolvR}_t^{-1}$). The solvency formulation is preferred throughout this study because as reserves shrink toward zero, the leverage ratio explodes and distorts the statistical distribution, whereas the Solvency Ratio converges smoothly to zero, yielding a well-behaved index for threshold estimation.

Fractional Reserve Theory (FRT) and loanable-funds: the central bank exogenously fixes the monetary base (H_t), which mechanically dictates bank lending and broad money through a stable fractional-reserve multiplier ($H \rightarrow \text{Deposits} \rightarrow \text{Loans} \rightarrow P$). In sharp contrast, Post-Keynesian Endogenous Money Theory (EMT) and Marxian monetary circuits establish that real-world banking operates in reverse causal sequence (Kaldor, 1982; Moore, 1988; Lavoie, 2014):

$$\Delta \text{Bank Loans} \longrightarrow \Delta \text{Deposits} \longrightarrow \Delta \text{Central Bank Reserves}(H) \quad (4)$$

Commercial banks extend working-capital loans to firms to finance payrolls and production. These loans simultaneously create bank deposits. The central bank, far from controlling the money supply as an exogenous policy instrument, is compelled to accommodate required reserves ex-post at the policy discount rate to maintain clearance in the interbank payments system. As Kaldor (1982) argued, credit money has no independent supply function: it comes into existence through the demand for bank lending and is extinguished through repayment. The central bank must accommodate this process to prevent the collapse of the credit and payments system. Monetarism, in this light, is not a neutral technical rule but a political strategy that deploys high interest rates and monetary contraction to induce unemployment and break trade union bargaining power. Under EMT, the traditional money multiplier ($mm_t \equiv M_t/H_t$) is not an exogenous causal driver but an unstable, ex-post accounting ratio reflecting commercial bank liquidity preferences, cash-drain ratios, and portfolio balance constraints. In peripheral capitalism, credit supply is structuralist and upward-sloping (Palley, 2013; Dow, 2006; Rochon, 1999): commercial banks' willingness to intermediate credit is bounded by external liquidity constraints and foreign exchange exposure.

3.11 This institutional reality establishes what I define as the *Peripheral Paradox*. In core economies issuing world money, the central bank accommodates domestic credit without facing an absolute currency boundary. In the periphery, however, while domestic credit generation is naturally endogenous ($\text{Loans} \rightarrow M_t$), the ultimate means of international settlement is bound to an exogenously constrained reserve asset ($e_t \cdot IR_t$). The peripheral central bank must back an endogenously expanding domestic credit pyramid with an externally constrained stock of world money. When external credit embargoes, copper price collapses, or private capital flight drain foreign reserves, the state cannot act as an unconstrained lender of last resort in hard currency; it must choose between contracting domestic activity or suffering balance-sheet insolvency. Under the UP, the nationalized commercial banking system under the State Development Corporation (CORFO) extended defensive working-capital credit to socialized and private enterprises facing distribution bottlenecks and price controls, compelling the Central Bank to monetize enterprise deficits. Domestic money growth thus functioned as an endogenous accommodation valve, driven by structural cost-push pressures rather than discretionary printing press impulses.

3.5 Financial Fragility Reinterpreted: State-Dependent Solvency of the Peripheral Central Bank

3.12 To understand how this balance-sheet contradiction generates macroeconomic crisis, I draw on the Financial Instability Hypothesis (FIH) of Minsky (1986). In Minsky's canonical framework, financial fragility is endogenously generated during prolonged expansions: stability breeds instability as private firms and banks voluntarily increase leverage, transitioning across qualitatively distinct financial positions (Kregel, 2008). However, Minsky formulated this dynamic for private actors in a closed, metropolitan economy possessing complete monetary sovereignty, where an unconstrained central bank (the "Big Bank") and fiscal authority (the "Big Government") stabilize an intrinsic unstable system. The peripheral on the other hand, central bank is neither private nor unconstrained. It does not lever up out of speculative euphoria; its leverage is structurally imposed by the balance-of-payments constraint and exter-

nally amplified by center shocks. Minsky’s categorical typology provides a theoretically useful description of qualitatively distinct solvency states, but whether these categories map onto the peripheral central bank—and how many distinct regimes the data supports—is a question that must be resolved empirically, not assumed a priori.

3.13 Reinterpreting Minsky for the peripheral central bank requires two analytical moves. First, [Foley \(2003, p. 21\)](#) demonstrates that during acute financial fragility, the state and the central bank as lender of last resort are compelled to absorb the insolvency of private enterprises onto their own balance sheets. The central bank does not lever up out of profit-seeking; it does so by executing its institutional mandate to prevent payments gridlock and economic breakdown. Second, [Chamorro-Futinicó \(2025b\)](#) demonstrates that financial instability in peripheral Latin American economies operates as an *externally amplified endogenous fragility*. Minsky’s dynamic cannot be applied to dependent formations without recognizing that external shocks from the center—the collapse of world monetary arrangements, terms-of-trade declines, and credit embargoes—serve as triggers detonating domestic balance-sheet fragility. In periods of financial insolvency, either by endogenous causes or exogenous political agency against an specific political process, the peripheral Central Bank enters a state of *forced defensive accommodation* not out of speculative euphoria, but under severe institutional confinement. When imperial sanctions cut off commercial trade credit, refusing domestic liquidity expansion would cause immediate real-sector collapse. Accommodating the deficits of socialized enterprises is an institutional imperative to maintain societal reproduction.

3.14 This reinterpretation establishes the central theoretical hypothesis of the chapter: the macroeconomic transmission behavior of the peripheral central bank is *state-dependent on its balance-sheet solvency*. The law of motion of the Solvency Ratio (Equation 3) implies that when $\text{Solv}R_t$ is sufficiently high, the central bank possesses genuine policy autonomy—domestic credit expansion mobilizes idle capacity without destabilizing the exchange rate, and external price shocks are absorbed at the border with minimal domestic pass-through. As solvency deteriorates past critical thresholds—driven by current account hemorrhage ($CA_t < 0$), capital flight ($KA_t < 0$), or accelerating domestic money growth ($g_{M,t}$)—the central bank loses this autonomy: forced defensive emission, lacking world-money backing, passes directly into price instability and currency flight. The empirical identification of where these thresholds lie, how many distinct regimes the data supports, and how transmission elasticities shift across solvency states is the object of the Threshold Vector Autoregression (TVAR) developed in Section ??.

4 Data Sources and Variable Construction

4.1 Monthly Panel: Sources and Sample Windows

The empirical core is a monthly panel of $N = 250$ observations (1960:01–1980:12), assembled programmatically from the Central Bank of Chile REST API (SieteRestWS). The API provides continuous monthly series for high-powered base money (H_t , thousands of pesos), the official nominal exchange rate (e_t , pesos per US dollar; pre-1975 escudo values retroactively converted at 1 peso = 1,000 escudos per the BCCh *Indicadores Económicos y Sociales de Chile 1960–2000* convention), London Metal Exchange copper cash prices (P_t^{Cu} , USD/lb), London bullion gold prices (P_t^{gold} , USD/oz), the US Wholesale Price Index ($P_{W,t}^{\text{US}}$), the official Consumer Price Index (π_t), and the US Dow Jones Industrial Average. The Santiago Stock Exchange (IGPA) index is drawn from the replication files of [Girardi and Bowles \(2018\)](#). Gross international reserves (IR_t , millions of USD) come from the IMF *International Financial Statistics*.

Sectoral real activity uses the disaggregated monthly component series compiled by [Díaz-Bahamonde \(2023\)](#): light industrial manufacturing (Q_t^{manuf}), copper mining extraction (Q_t^{mining}),

and physical export and import volumes (Exports_t , M_t). I use the individual components, not his aggregated IMEA composite index. The components are harmonized from *Estadística Chilena* (DGE/INE) and the BCCCh *Boletín Mensual* and are available from 1927 onward; I restrict the panel to 1960:01–1980:12 to align with the monetary and reserve series.

Narrow money ($M1_t$, thousands of pesos) and the observed banking multiplier ($m_t \equiv M1_t/H_t$) are available only over the un-spliced window 1966:01–1980:12 ($N = 180$), which excludes pre-1966 commercial bank classification revisions. $M1_t$ enters the empirical architecture exclusively in the commercial-banking bifurcation and multiplier deconstruction steps of the Granger causality tournament (Section ??); the primary solvency threshold and the full-panel TVAR use H_t throughout.

A supplementary annual tier ($T = 91$, 1920–2010) provides the long-run accumulation backdrop used in Sections 5.1 and 5.2: the functional wage share (ω_t) from Astorga (2023); the productive capital stock ($K_t \equiv K_t^{\text{ME}} + K_t^{\text{NRC}}$, excluding residential construction), the aggregate profit rate (r_t , nominal surplus on productive fixed capital at current replacement cost), and structural capacity utilization (μ_t , estimated via cointegrating polynomial regression) from the GPIM dataset of Chapter 2; and balance-of-payments accounts from the World Historical Balance of Payments Database (Nievas and Piketty, 2025). Appendix ?? catalogs complete series identifiers and metadata.

4.2 Variable Construction

The primary Central Bank reserve solvency ratio, constructed over the full monthly panel, is

$$\text{SolvR}_t^H \equiv \frac{e_t \cdot IR_t}{H_t \cdot 1000}, \quad (5)$$

where e_t is in pesos per dollar, IR_t in millions of dollars, and H_t in thousands of pesos. The factor of 1,000 converts H_t to millions of pesos, aligning the denominator with the domestic-currency valuation of reserves ($e_t \cdot IR_t$). Base money is the appropriate primary liability because it is the Central Bank’s direct obligation and is observed continuously from 1960:01. In the threshold specification, the predetermined lag SolvR_{t-1}^H satisfies the consistency conditions of Hansen (1999).

For the commercial-banking decomposition ($N = 180$), I also construct

$$\text{SolvR}_t^{M1} \equiv \frac{e_t \cdot IR_t}{M1_t \cdot 1000},$$

which correlates with SolvR_t^H at $\rho = 0.974$. This measure is not the primary solvency variable; it is used only to test whether the commercial credit channel alters the ordering between external backing and domestic liabilities.

The Prebisch-ECLAC capacity to import measures the purchasing power of physical exports relative to imported manufactured inputs:

$$M_t^{\text{cap}} \equiv \left(\frac{\text{Exports}_t}{\text{Exports}_{1970:11}} \right) \times \left[\frac{P_t^{\text{Cu}}/P_{1970:11}^{\text{Cu}}}{P_{W,t}^{\text{US}}/P_{W,1970:11}^{\text{US}}} \right] \times 100, \quad (6)$$

benchmarked to November 1970 (1970:11 = 100). The real structural imbalance ratio is

$$\Theta_t \equiv \left(\frac{M_t}{M_t^{\text{cap}}} \right) \times 100. \quad (7)$$

When $\Theta_t > 100\%$, physical import volumes exceed export purchasing power, generating a foreign-exchange deficit that must be financed through credit lines or reserve depletion.

4.3 Data Adjustments

The raw customs import volume series exhibits a single-month administrative registration drop in April 1973: the recorded volume falls from 2.88 (March) to 0.26 (April), recovering to 2.63 (May). Historical verification confirms this reflects a paperwork lag in clearing port arrivals into official records, not a collapse in physical trade. I replace the April observation with the monotonic linear midpoint, 2.76, to prevent the registration artifact from triggering a spurious structural break in the VAR systems.

4.4 Integration Order of Variables Included in Empirical Exercises

The transformed series entering the Toda–Yamamoto and threshold VAR architectures are stationary in levels. Table 1 reports the ADF diagnostics; the underlying level series have maximal integration order $d_{\max} = 1$.

Table 1: Augmented Dickey–Fuller Unit Root Tests

Variable	Levels (z_t)		First Difference (Δz_t)		Order
	t -ADF	5% CV	t -ADF	5% CV	
CPI Inflation (π_t)	-3.372	-2.87	-13.619***	-2.87	$I(0)$
Base Money Growth ($g_{H,t}$)	-3.751	-2.87	-12.484***	-2.88	$I(0)$
Narrow Money Growth ($g_{M1,t}$)	-4.059	-2.88	-17.812***	-2.88	$I(0)$
Multiplier Growth ($\Delta \ln m_t$)	-12.791	-2.88	-11.957***	-2.88	$I(0)$
Manufacturing Output ($g_{\text{Manuf},t}$)	-14.105	-2.87	-19.409***	-2.88	$I(0)$
Mining Extraction ($g_{\text{Mining},t}$)	-14.910	-2.87	-13.947***	-2.88	$I(0)$
Structural Imbalance ($\Delta \ln \Theta_t$)	-14.512	-2.87	-17.171***	-2.88	$I(0)$
World Gold Price Growth ($g_{P,\text{gold},t}$)	-10.559	-2.87	-17.959***	-2.88	$I(0)$
Nominal Exchange Rate Growth ($g_{e,t}$)	-3.777	-2.87	-21.142***	-2.88	$I(0)$
Solvency Ratio Growth ($\Delta \ln \text{SolvR}_t^H$)	-8.726	-2.87	-14.011***	-2.88	$I(0)$

Notes: Monthly series, 1960:01–1980:12 ($N = 250$), except narrow money and multiplier, 1966:01–1980:12 ($N = 180$). ADF tests include constant drift; lag orders selected via SBIC ($p \leq 4$). MacKinnon (1996) finite-sample critical values. *** $p < 0.01$.

5 Historical Background, Stylized Facts, and Empirical Results

5.1 Historical Background: The Rise of the Unidad Popular

The Unidad Popular’s electoral victory in 1970 brought a project of socialist transformation into contact with popular organizations already contesting authority over work, land, and urban life. Industrial workers, the peasantry, and *pobladores* did not enter politics through the same institutions or pursue identical demands. Their struggles nevertheless became increasingly connected before the election. This subsection follows successive waves of mobilization, repression, and political articulation to explain that connection. Chile’s changing economy supplied the structural setting, but did not dictate the political outcome.

During the nitrate era, export receipts sustained public revenues and domestic activity while tying employment, imports, and fiscal capacity to external demand and finance. The postwar dislocation of natural nitrate weakened that arrangement before the Great Depression. The Depression intensified an existing crisis of the export economy and accelerated protection of domestic production and state intervention. Manufacturing development and the institutional

conditions for industrialization thus preceded 1929; the crisis changed their scale and direction rather than initiating them from nothing (Pinto Santa Cruz, 1958; Palma, 1984; Silva, 2007).

After the Second World War, mass production, productivity growth, rising wages, collective bargaining, and mass consumption became linked in the North Atlantic core (Vidal, 2019). US-centered monetary and political leadership reorganized international hierarchy without removing the unequal conditions under which peripheral economies industrialized. Chile expanded production for its domestic market while continuing to require imported machinery, intermediate inputs, technology, and foreign exchange. Industrial growth could therefore reproduce external dependence in a new form (De Vylder, 1976; Silva, 2007).

Industrialization also changed the spatial distribution of labor. Figure 1 places declining primary-sector shares beside rising urbanization and tertiary activity. The secondary sector did not become the principal absorber of the expanding labor force as Chile urbanized. This relative shortfall supports the characterization of *disarticulated industrialization*: industrial expansion coexisted with a wider transformation of settlement and livelihoods (De Vylder, 1976). The conflicts that developed on this terrain, rather than the sectoral series themselves, establish the chronology that follows.

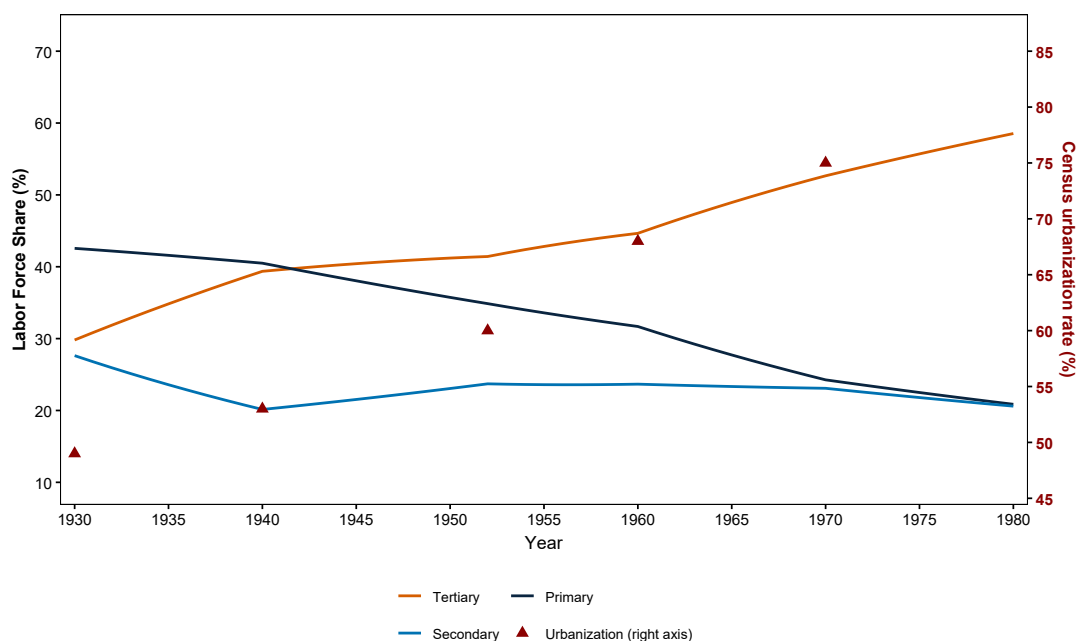


Figure 1: Sectoral labor-force composition and urbanization in Chile, 1930–1980.

Source: Author's elaboration from Clio-Lab PUC historical statistics (Díaz et al., 2016). Sectoral series refer to the labor force; urbanization refers to population. These measures have different denominators. Census urbanization observations are plotted as points.

Wave I: Selective incorporation and coercive backlash. The first conflict wave began before the profitability series available here starts in 1943. Popular Front politics brought parties and organized constituencies into a common electoral project, while access to collective representation remained uneven across workplaces and the countryside (Milos, 2008; Loveman, 1976). Rural workers made claims over employment, labor conditions, and the right to organize; the countryside was not merely a passive source of labor for urban industry. The 1943–47 accounting interval captures an observed economic segment of this longer confrontation, not its historical origin.

By the mid-1940s, expanding rural demands formed part of a wider confrontation over labor

organization and the terms of work. The Plaza Bulnes massacre and the national stoppages of January and February 1946 exposed its urban and industrial dimensions (Bravo Vargas, 2017, pp. 199–201, 213–225). The agricultural petitions assembled by Loveman document the rural dimension and its crest in 1947 (Loveman, 1976). Their distributive significance lay partly in workers’ capacity to turn shared grievances into demands extending beyond an individual workplace. Over the observed 1943–47 interval, a declining profit share was the principal negative contribution to profit-rate growth in the chapter’s accounting (Table 2).

The response in the countryside combined formal recognition with restrictions on the scale of organization. Law 8,811, published in July 1947, recognized agricultural unions while confining them to individual *fundos* and prohibiting combinations of those unions. Eligibility requirements tied organization to a sufficiently large and stable workforce on each property; restrictions on petition filing during sowing and harvest and the resort to binding arbitration further narrowed collective action (República de Chile, 1947, arts. 9, 14, 16, 53, 60–69). The law made the boundaries of landed estates the boundaries of legal representation, obstructing the organization of common agrarian demands across them. Recognition carried material benefits and limited protections, but preserved constraints on the pursuit of collective claims.

Law 8,987, published in September 1948, extended restriction through political exclusion. Alongside outlawing the Communist Party, it connected exclusion from the electoral registers to exclusion from union membership and barred affected workers from union leadership, labor delegations, and conciliation bodies, including agricultural arbitration boards (República de Chile, 1948, art. 1, amendment 5; art. 7, amendment 1; art. 8). The two laws restricted different levels of organization: combination across estates and political representation within a broader national confrontation. The subsequent decline in registered agricultural petitions marks the contraction of these channels, not the disappearance of agrarian grievances.

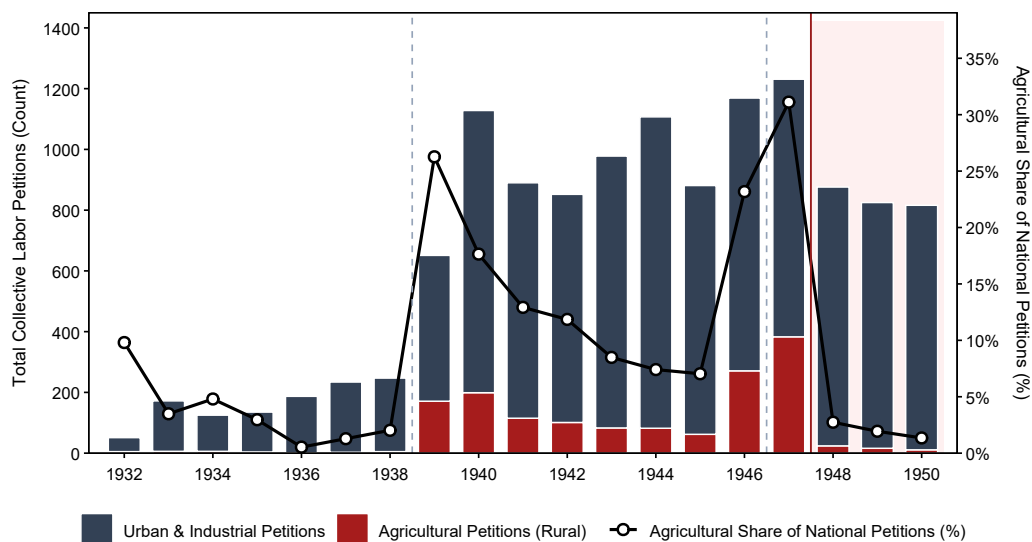


Figure 2: Recorded collective labor petitions by sector, 1932–1950.

Source: Author’s archival compilation from Loveman (1976). Petitions measure registered collective demands under the prevailing legal framework, not all forms of social conflict.

Wave II: Coercive containment and urban recomposition. The restrictions imposed in 1947–48 narrowed the institutional channels through which workers could press their demands. In the countryside, legal recognition bound union organization to the estate; across sectors, anticommunist exclusion attacked the political networks through which collective claims acquired a wider reach. Conflict continued under conditions increasingly unfavorable to workers’ efforts

to defend wages and influence economic policy (Thielemann Hernández, 2023, pp. 47–53).

Stabilization brought this conflict over distribution into sharper focus. Under Ibáñez, the US advisory firm Klein–Saks supplied a technical rationale for policies restraining wage adjustment in the name of controlling inflation. As Thielemann Hernández (2023, pp. 45–53) argues, these measures consolidated a deterioration in real wages already underway rather than initiating it. Limits on wage adjustment, unemployment, and repression of organized resistance shifted adjustment costs onto working-class households, while rising prices for essential goods and services extended the dispute beyond the workplace.

The profitability accounts place this distributive conflict alongside a continuing transformation of production. Over 1948–58, the rising profit share supplied the largest positive contribution to profit-rate growth. Capital deepening more than offset the positive contribution of potential labor productivity: their annualized contributions were –2.60 and +2.29 log points, respectively (Table 2). Capital per worker expanded faster than potential output per worker, while distribution supported profitability. This greater capital intensity coexisted with the limited relative absorption of the growing labor force described above.

Urban growth made housing central to the conflict over living standards. A place in the city required more than access to wage employment: households confronted scarce housing, high rents, inadequate services, and the cost of reaching work. These pressures cut across the division between securely employed workers and residents with irregular incomes. Thielemann traces how disputes over transport fares, rents, and basic consumption connected grievances at work with organization in neighborhoods. The city became a field of collective action in its own right (Thielemann Hernández, 2023, pp. 51–53).

The revolt of April 1957 and the occupation that founded La Victoria in October gave this urban conflict a new political expression. They were distinct forms of action, but Thielemann places them within a shared process through which popular groups challenged the limits of institutional representation and expanded their capacity to act collectively. La Victoria made occupation and neighborhood construction into a practical claim to housing. The significance of 1957 lies in this political awakening rather than an immediate reversal of labor’s economic position. It opened a phase of struggle whose organizations, alliances, and objectives developed unevenly through the Long Sixties (Thielemann Hernández, 2023).

Wave III: The Long Sixties and the rise of the Unidad Popular. Industrial workers entered the Long Sixties with demands reaching beyond wage adjustment to the organization of production. Disputes over rationalization and work intensity challenged managerial authority, while solidarity linked workers across industries: Huachipato steelworkers supported the Lota miners in the long strike of 1960. Repression shaped collective memory, including the killings at El Salvador in 1966. These experiences helped radicalize sections of the movement before the UP took office (Thielemann, 2018, pp. 119–125).

In the countryside, agrarian reform placed landownership and the organization of rural labor within the same political struggle (Loveman, 1976). The 1967 union law changed the territorial basis on which workers could act. Membership could extend across estates, the minimum territorial unit became the commune, and unions could form federations and confederations (República de Chile, 1967, arts. 1, 3–4). Compared with the estate-bound framework of 1947, this widened the institutional scope for collective demands. The agrarian question concerned both access to land and the power to organize across the boundaries of individual properties.

In the cities, *pobladores* pressed claims to land, housing, and public services through neighborhood organization and occupations. Their mobilization challenged who could secure a place in the city and on what terms. Housing was both a material need and a demand for political

recognition, bringing everyday life into conflicts over property and public provision ([Garcés, 2002](#); [Murphy, 2015](#)). These struggles had their own organizations and objectives, even where their participants also experienced conflict as wage earners.

Concepción illustrates the local relationships through which political connections developed. Schlotterbeck locates organizing in the university, nearby industrial communities, and working-class neighborhoods, where labor and student activism intersected with struggles over housing. These relationships preceded Allende's victory, while their development under the UP opened further possibilities for coordinated action ([Schlotterbeck, 2018](#), pp. 7, 11–14). Popular articulation was a process, not a unity completed at the ballot box.

Widening mobilization did not coincide with a continuous deterioration of aggregate profitability. Across the 1959–70 accounting interval, the profit rate rose, supported by positive contributions from the profit share, utilization, and potential labor productivity, with capital deepening providing an offset (Table 2). The conflict concerned the distribution and control of an expanding economy as well as its unresolved limits. By 1970, struggles over work, land, and housing confronted the question of executive power; their convergence made a common political project possible without erasing differences over its objectives or means.

Wave IV: The UP project and the crisis of accumulation. Allende's election in 1970 brought these accumulated struggles into a new relationship with executive power. The UP's project connected redistribution to changes in ownership and the direction of production. In Arboleda's account, Pedro Vuskovic and Jacques Chonchol approached planning as part of national liberation: transforming the economy required both public intervention and greater popular power ([Arboleda, 2023](#)). The government encountered organizations with their own demands and rhythms of action. Winn's distinction between revolution from above and below captures this relationship between a state project and popular initiatives that could support it while pressing beyond its immediate limits ([Winn, 1986, 2016](#)).

Political articulation continued after the election. In Concepción, the People's Assembly of 1972 brought together sectors whose organizing Schlotterbeck traces through the preceding years ([Schlotterbeck, 2018](#), p. 14). This local convergence illustrates why the rise of the UP cannot be reduced to its electoral coalition: participation also developed through institutions and practices created within the movements themselves.

These transformations unfolded alongside a sharp deterioration in profitability and productive accumulation. In the chapter's accounts, the profit rate fell from 12.23 percent in 1970 to 6.57 percent in 1972, while net accumulation slowed from 4.23 to 1.85 percent. Gross accumulation fell from 11.63 to 7.64 percent over the same years (Table 3). The annual observations for 1973 combine a profit-rate rebound with a further slowing of net accumulation to 1.71 percent; they span both the final months of the UP and the first months after the coup. This sequence directs attention from profitability alone to the conditions under which surplus was converted into productive investment.

The confrontation over property was also visible in financial markets: share prices fell sharply after Allende's election and rose after the military coup ([Girardi and Bowles, 2018](#)). Productive accumulation, however, remained weak in the annual accounts. The coup of September 1973 ended the UP experiment through force, not through the automatic resolution of an economic adjustment. The following analysis examines how distribution, utilization, relative prices, and external constraints interacted with the deterioration of profitability and accumulation during the crisis.

Table 2: Social Conflict Waves and Profitability Decomposition, Chile, 1943–1975

Wave of Social Conflict	Profit Rate $\Delta \ln r$	Profit Sh. $\Delta \ln(1 - \omega)$	Rel. Price $\Delta \ln(P_Y/P_K)$	Cap. Util. $\Delta \ln \mu$	Lab. Prod. $\Delta \ln y^p$	Cap. Deep. $-\Delta \ln k$
<i>A. Social Conflict Waves (Annualized Rates and % Shares)</i>						
Wave I: Observed accounting segment (1943–1947)	-2.84%	-5.53% (+194.9%)	+2.63% (-92.7%)	+0.62% (-21.7%)	-0.21% (+7.6%)	-0.34% (+11.9%)
Wave II: Containment and urban recomposition (1948–1958)	+2.93%	+3.11% (+106.1%)	-0.10% (-3.3%)	+0.23% (+7.9%)	+2.29% (+77.9%)	-2.60% (-88.6%)
Wave III: Long Sixties accounting interval (1959–1970)	+2.07%	+1.31% (+63.3%)	+0.45% (+21.6%)	+0.97% (+46.9%)	+1.72% (+83.0%)	-2.38% (-114.8%)
Wave IV: UP crisis accounting interval (1971–1973)	-7.65%	+1.94% (-25.3%)	-4.35% (+56.9%)	-4.78% (+62.5%)	-1.23% (+16.0%)	+0.77% (-10.1%)
<i>B. Terminal Crisis Cycle: Annual Log Changes and % Shares</i>						
1970 → 1971: Reactivation peak	-14.05%	-11.44% (+81.4%)	-7.90% (+56.2%)	+5.92% (-42.1%)	-0.43% (+3.1%)	-0.20% (+1.4%)
1971 → 1972: Profit squeeze	-48.08%	-34.60% (+72.0%)	-10.42% (+21.7%)	-2.52% (+5.2%)	-2.27% (+4.7%)	+1.74% (-3.6%)
1972 → 1973: Annual observation spanning the coup	+32.78%	+38.48% (+117.4%)	+1.72% (+5.3%)	-7.04% (-21.5%)	-0.18% (-0.5%)	-0.20% (-0.6%)
1973 → 1974: DL 522 compression	+15.74%	+28.59% (+181.7%)	-11.47% (-72.9%)	-0.72% (-4.6%)	+4.04% (+25.7%)	-4.70% (-29.9%)
1974 → 1975: Chicago shock therapy	-48.97%	-17.70% (+36.2%)	-16.24% (+33.2%)	-14.78% (+30.2%)	+8.97% (-18.3%)	-9.22% (+18.8%)

Notes. Growth-accounting decomposition using the identity $\Delta \ln r_t \equiv \Delta \ln(1 - \omega_t) + \Delta \ln(P_{Y,t}/P_{K,t}) + \Delta \ln \mu_t + \Delta \ln y_t^p - \Delta \ln k_t$. Values outside parentheses are annualized log contributions (% p.a.). Values inside parentheses report the component's share of the total change, computed as $(\Delta \ln X_i / \Delta \ln r_t) \times 100\%$. A positive share indicates the component moved in the same direction as the total change; a negative share indicates it offset the total change. Shares may exceed 100% due to normalization by the net total change. Source: Author's own elaboration (Chapter 2).

Table 3: Capital Accumulation Response Across the Terminal Crisis Cycle, Chile, 1964–1975

Episode / Year	Profit Rate r (%)	Accum. Propensity χ (ratio)	Gross Acc. Rate g_K^{gross} (%)	Depreciation Rate δ (%)	Net Acc. Rate g_K^n (%)
<i>A. Christian Democratic Administration (1964–1970)</i>					
1964 (Baseline)	8.16	1.530	12.49	7.67	4.82
1965	8.59	1.244	10.69	6.94	3.75
1966	10.94	0.972	10.63	6.97	3.66
1967	10.14	1.066	10.81	7.06	3.75
1968	11.29	0.991	11.20	7.24	3.96
1969 (Peak Profit.)	12.79	0.857	10.96	7.14	3.82
1970 (Inherited)	12.23	0.951	11.63	7.40	4.23
<i>B. Unidad Popular Administration (1971–1973)</i>					
1971 (Reactivation)	10.62	0.953	10.12	6.78	3.34
1972 (Squeeze)	6.57	1.162	7.64	5.78	1.85
1973 (Coup year)	9.12	0.821	7.48	5.78	1.71
<i>C. Military Dictatorship & Shock Therapy (1974–1975)</i>					
1974 (Compression)	10.67	0.811	8.65	6.27	2.38
1975 (Collapse)	6.54	1.035	6.77	5.56	1.21

Notes. Accumulation accounting for productive fixed assets ($K_t \equiv K_{\text{ME},t} + K_{\text{NRC},t}$), excluding residential construction. Gross accumulation is $g_{K,t}^{\text{gross}} \equiv I_t^{\text{gross}}/K_{t-1}$; physical depreciation is $\delta_t \equiv D_t/K_{t-1}$; net accumulation is $g_{K,t}^n \equiv I_t^{\text{net}}/K_{t-1}$; and the accumulation propensity out of gross surplus is $\chi_t \equiv g_{K,t}^{\text{gross}}/r_t$. All observations satisfy the exact identity $g_{K,t}^n \equiv \chi_t r_t - \delta_t$, subject to displayed rounding. The profit rate is a level, not its growth rate. Annual 1973 observations span the coup. Source: Author's own elaboration (Chapter 2).

5.2 Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints

5.3 Empirical Exercise: *Lakatosian* Granger Causality Tournament

5.3.1 Econometric Framework: Toda–Yamamoto Levels Augmentation

5.3.2 Univariate Stationarity and Integration Pre-testing Battery ($d_{\max} = 1$)

5.3.3 Sequential Tournament Architecture and Hypothesis Testing Rules

5.3.4 Empirical Findings

5.3.5 Residual Diagnostics and Dynamic Stability

5.4 Empirical Exercise: Non-Linear Dynamics under Peripheral Central Bank Solvency. A Threshold VAR Approach

5.4.1 The Threshold VAR Specification and Structural Identification

5.4.2 Endogenous Threshold Estimation and Residual Bootstrap Linearity Testing

5.4.3 Regime Parameter Estimates and System Stability Forensics

5.4.4 Dynamic Generalized Impulse Response Functions (GIRFs)

5.4.5 Formal Multiplier Difference Hypothesis Tests

6 Political Economy Discussion, Global Implications, and Concluding Synthesis

6.1 The Political Economy of Peripheral Solvency and Subordinated Money

5.1 The econometric findings in Section 5 dismantle the foundational premise shared by orthodox interpretations of the Unidad Popular: the assumption that money growth is an unconstrained, exogenous policy choice. By showing that inflation and currency depreciation systematically preceded base money expansion ($\pi_{t-1} \rightarrow g_{M,t}$), and that the inflationary pass-through of monetary growth is strictly state-dependent on central bank reserve solvency (SolvR_t), my analysis opens the “black box” of peripheral monetary transmission. Central Bank money creation operated as the institutional accommodation valve through which the monetary authority, acting as lender of last resort (LLR), mediated the collision between domestic distributive commitments and an externally enforced foreign-exchange blockade.

5.2 In a peripheral capitalist economy, the central bank operates under what I define as the *Peripheral Paradox* (Section ??). While the domestic credit pyramid is naturally endogenous—commercial lending creates deposits, which the central bank must accommodate to prevent liquidity gridlock—the ultimate means of international settlement is exogenously rationed by access to world money ($e_t \cdot IR_t$). When external commercial credit lines are severed and copper receipts are tied up in litigation, the peripheral central bank cannot freely print international purchasing power. As reserves drain, the domestic monetary base (H_t) becomes structurally decoupled from foreign exchange backing. The central bank is placed in an impossible insti-

tutional bind: either it refuses liquidity to state-owned enterprises and private firms facing import cost explosions, triggering immediate mass bankruptcies and physical output collapse; or it acts as the lender of last resort, injecting emergency liquidity that, in the absence of foreign reserves to anchor the currency, accelerates domestic hyperinflation. The descent of the central bank’s own balance sheet into a Forced Ponzi regime ($\text{SolvR}_m \leq 0.14$), transposing the Minsky-Foley framework (Foley, 2003; Chamorro-Futinico, 2025b) to peripheral central bank solvency, registered the structural exhaustion of peripheral solvency under external financial subordination.

5.3 This institutional reality exposes the flaw in the orthodox treatment of retail shortages, rationing, and black markets. In the monetarist narrative of Edwards (2024), shortages are treated as unambiguous evidence of “repressed inflation” caused by price controls. In structural causal terms, however, retail shortages (C) constitute an *endogenous collider variable* (Pearl, 2009) caused jointly by wage-led demand stimulus (U_1) and private wholesale distribution sabotage, trucking strikes, and hoarding (U_2):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp U_2 \mid C \quad (8)$$

By conditioning on empty store shelves without modeling the October 1972 Bosses’ Lockout (*Paro de Octubre*) or the deliberate withholding of inventories by the wholesale distribution monopoly (SDI), orthodox accounts commit an M-bias collider stratification error. They mistake the physical symptoms of organized class struggle and distribution sabotage for the mechanical consequences of central bank balance sheets, generating a powerful causal illusion that conceals the political economy of destabilization (Blanco and Matute, 2018; Casals, 2021).

6.2 The Unidad Popular in the Global Structural Crisis of the 1970s

5.4 The crisis of the Unidad Popular cannot be understood in national isolation. It was the peripheral expression of a global structural crisis of capitalism that unfolded across the early 1970s. The long postwar boom in the advanced capitalist core was exhausting itself: corporate profit rates in the United States and Western Europe suffered a protracted secular decline (Weiskopf, 1979; Bowles et al., 1983), labor militancy intensified, and the institutional foundations of Atlantic Fordism began to fracture. When President Richard Nixon closed the gold window on August 15, 1971, he unilaterally dissolved the gold-dollar convertibility anchor of the Bretton Woods system. This hegemonic shock exported U.S. stagflation directly to the periphery while reconfiguring the international currency hierarchy.

5.5 For Chile, the timing was catastrophic. The Vuskovic economic program had been designed in late 1970 under the explicit assumption of a stable, fixed-exchange-rate international monetary regime backed by ample inherited reserves. Nine months into the administration, the institutional anchor of world money vanished. U.S. inflation drove up the dollar prices of Chilean imports, while the Nixon administration’s directive to “make the economy scream” mobilized multilateral lending freezes (World Bank, IDB, Eximbank) and private commercial credit boycotts. The hegemonic state, facing its own domestic profitability crisis, deployed external financial subordination as an offensive instrument of imperial discipline against a peripheral nation exercising sovereignty over its natural resources (Harmer, 2011; Arboleda, 2023). The Unidad Popular fell when its redistributive commitments collided directly with the hard boundaries of the international currency hierarchy during an acute global structural crisis.

6.3 Policy Implications for Contemporary Peripheral Sovereignty

5.6 This historical reassessment yields fundamental analytical and policy lessons for contemporary peripheral formations attempting progressive or socialist developmental transitions. The

Chilean experience provides a definitive critique of naive extensions of Modern Monetary Theory (MMT) to the global South. In an asymmetric international monetary hierarchy, sovereign authority over domestic currency issue does not confer command over world money (Vasudevan, 2009; Alami et al., 2023). A peripheral government cannot simply finance developmental programs through central bank monetization without confronting the structural ceiling of foreign-exchange reserve solvency.

5.7 Peripheral monetary autonomy is strictly conditioned by the width of the external corridor. Sustaining developmental transitions requires active macroprudential management of foreign exchange, including mandatory surrender of export revenues, differentiated exchange-rate bands, and permanent, counter-cyclical capital controls to insulate domestic policy from international liquidity cycles (Ffrench-Davis, 2018; Vasudevan, 2019). High nominal commodity export revenues are also insufficient to secure development if the physical composition of imports is vulnerable to external leverage. Peripheral planning must prioritize strategic self-sufficiency in non-substitutable capital equipment, replacement parts, and domestic food security to prevent foreign-exchange blockades from paralyzing the domestic productive apparatus.

6.4 Concluding Synthesis: Toward a Progressive Research Programme

5.8 For more than fifty years, the historiography of the Unidad Popular has operated as an ideological weapon—a cautionary tale constructed by monetarist and Austrian orthodoxy to declare the impossibility of socialist transformation. By subjecting these claims to a formal Neo-Lakatosian Hypothesis Tournament, this chapter demonstrates that the orthodox narrative collapses under econometric scrutiny. First, the Monetarist hypothesis (H_1) fails: prior price shocks unidirectionally forced emergency monetary accommodation on the central bank balance sheet. Second, the Austrian ABCT framework (H_2) fails: the 1971 output surge represented a Keynesian-Structuralist reactivation of idle industrial capacity ($u = 98.4\%$) alongside contracting fixed capital formation ($\Delta K = -2.3\%$), ruling out credit-driven malinvestment. Third, the Structural-Dependency thesis (H_3) holds: the inflationary pass-through of monetization is strictly governed by central bank solvency thresholds (SolvR_t). Domestic liquidity is absorbed without inflation when reserves are solvent ($\text{IRF}_\pi = 0.00\%$), but detonates into hyperinflation once imperial blockades and reserve exhaustion push the central bank balance sheet into the Forced Ponzi regime ($+4.54\%$ at $h = 0$).

5.9 Combining the relational thickness of a canonical case study with regime-switching econometrics, this chapter advances a *progressive problem shift* within the dependency research programme (Kvangraven, 2021; King and Sznajder, 2006). The analytical framework formalized here—the Solvency Ratio threshold model, the multi-frequency tournament architecture, and the decomposition of import composition constraints—provides a generative toolkit for analyzing peripheral financial subordination across historical and contemporary settings. From the structural crises of the Southern Cone in the 1970s and 1980s (Argentina, Peru) to contemporary debt traps in the global South, the central bank balance sheet remains the contested terrain where external financial subordination enters domestic reproduction. Reclaiming sovereignty in the periphery requires understanding that the struggle for economic democracy cannot stop at the borders of the nation-state; it must confront and transform the hierarchical architecture of the world monetary system.

References

- Abeles, M. (2013). Duncan Foley’s circuit of capital model for an open economy. In *Social Fairness and Economics*, pages 228–253. Routledge.
- Afonso, A., Baxa, J., and Slavík, M. (2018). Fiscal developments and financial stress: Does financial stress matter for fiscal policy? *Empirical Economics*, 55(4):1851–1882.
- Ahumada, J. M. (2019). *Political Economy of Peripheral Growth*. Springer.
- Ahumada, J. M. and Chang, H.-J. (2025). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Ahumada, J. M. and Torres, M. (2024). Dependency theory: an underdevelopmental thought from latin america to the entire world. In *Global Handbook of Inequality*, pages 1–21. Springer.
- Alami, I., Alves, C., Bonizzi, B., Kaltenbrunner, A., Koddenbrock, K., Kvangraven, I., and Powell, J. (2023). International financial subordination: a critical research agenda. *Review of International Political Economy*, 30(4):1360–1386.
- Aldunate, F., González, F., and Prem, M. (2022). The effects of U.S. banking covert action during Allende’s Chile. *Journal of Financial Economics*. GAP FLAG: confirm journal, volume, pages.
- Arboleda, M. (2023). Development as national liberation: The experience of the Popular Unity government in Chile. *Radical Philosophy*, (215):23–38.
- Astorga, P. (2023). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Balke, N. S. (2000). Credit and economic activity: credit regimes and nonlinear propagation of shocks. *Review of Economics and Statistics*, 82(2):344–349.
- Bambirra, V. (1971). *Tipología de la dependencia en América Latina*. CESO, Universidad de Chile, Santiago.
- Baran, P. A. (1957). *The political economy of growth*. Monthly Review Press, New York.
- Basu, D. (2022). *The Logic of Capital: An Introduction to Marxist Economic Theory*. Cambridge University Press, Cambridge.
- Bautista, M., González, F., Martínez, M., Muñoz, P., and Prem, M. (2023). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Bitar, S. (2013). *Chile 1970–1973: asumir la historia para construir el futuro*. Peña Lillo / Ediciones América del Sur, Santiago. Originally published 1979.
- Blanco, F. and Matute, H. (2018). The illusion of causality: a cognitive bias underlying pseudoscience. In Kaufman, A. B. and Kaufman, J. C., editors, *Pseudoscience: The Conspiracy Against Science*, pages 45–76. MIT Press, Cambridge, MA.
- Bockman, J. (2019). Democratic socialism in chile and peru: Revisiting the “chicago boys” as the origin of neoliberalism. *Comparative Studies in Society and History*, 61(3):654–679.
- Bowles, S., Gordon, D. M., and Weisskopf, T. E. (1983). *Beyond the Waste Land: A Democratic Alternative to Economic Decline*. Anchor Press/Doubleday, Garden City, NY.
- Braun, J., Braun, M., Briones, I., Díaz, J., Lüders, R., and Wagner, G. (2000). Economía chilena 1810–1995: Estadísticas históricas. Documento de Trabajo 187, Instituto de Economía, Pontificia Universidad Católica de Chile.
- Bravo Vargas, V. (2017). Chile no va hoy a la fábrica: Protesta obrera y represión política en el verano de 1946. *Izquierdas*, 35:199–232.
- Caputo, O. and Pizarro, R. (1970). *Imperialismo, dependencia y relaciones económicas internacionales*. CESO, Universidad de Chile, Santiago.

- Cárdenas, E., Lana, L., and Seabra, R. (2022). *El CESO y la dependencia: orígenes, debates y legado*. CLACSO. GAP FLAG: confirm publisher and editors.
- Cardoso, F. H. and Faletto, E. (1979). *Dependency and development in Latin America*. University of California Press, Berkeley. Originally published in Spanish, 1969.
- Casals, M. (2021). The insurrection of the middle class: Social mobilization and counterrevolution during the popular unity government, chile, 1970–1973. *Journal of Social History*, 54(3):944–969.
- Céspedes, L. F. and Chang, R. (2024). Optimal foreign reserves and central bank policy under financial stress. *American Economic Journal: Macroeconomics*, 16(3):230–267.
- Chamorro-Futinico, J. C. (2025a). Boom and bust: Dynamics of systemic financial instability in financialized emerging economies. *Review of Political Economy*, 37(5):1757–1788.
- Chamorro-Futinico, J. C. (2025b). Boom and bust: Dynamics of systemic financial instability in financialized emerging economies. *Review of Political Economy*, Ahead-of-print:1–25.
- Charnysh, V., Finkel, E., and Gehlbach, S. (2023). Historical political economy: core concepts and new directions. In Jenkins, J. A. and Rubin, J., editors, *The Oxford Handbook of Historical Political Economy*. Oxford University Press. GAP FLAG: confirm if this is a standalone chapter or part of handbook introduction; verify year 2023 vs 2024.
- Clark, T. D. (2017). Rethinking Chile’s “Chicago Boys”: neoliberal technocrats or revolutionary vanguard? *Third World Quarterly*, 38(6):1350–1365.
- Cuesta, J. I., Gallego, F., and González, F. (2015). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- De Castro, S. (1992). *El Ladrillo: bases de la política económica del gobierno militar chileno*. Centro de Estudios Públicos, Santiago. Originally written 1973; published 1992.
- De Vylder, S. (1976). *Allende’s Chile: the political economy of the rise and fall of the Unidad Popular*. Cambridge University Press, Cambridge.
- Díaz, J., Lüders, R., and Wagner, G. (2016). *Chile 1810–2010: La República en Cifras*. Ediciones Universidad Católica de Chile, Santiago, Chile.
- Díaz-Bahamonde, J. (2023). An index of monthly economic activity for chile, 1927–2021: A proposal. *Documento de Trabajo, Instituto de Economía UC*.
- Dornbusch, R. and Edwards, S. (1989). Macroeconomic populism in Latin America. NBER Working Paper 2986, National Bureau of Economic Research, Cambridge, MA.
- Dornbusch, R. and Edwards, S. (1990). *The Macroeconomics of Populism in Latin America*. University of Chicago Press.
- Dornbusch, R. and Edwards, S. (1991). The macroeconomics of populism. In Dornbusch, R. and Edwards, S., editors, *The Macroeconomics of Populism in Latin America*, pages 7–13. University of Chicago Press, Chicago.
- Dos Santos, T. (1970). The structure of dependence. *American Economic Review*, 60(2):231–236.
- Dow, S. C. (2006). Endogenous money: Structuralist. In Arestis, P. and Sawyer, M., editors, *A Handbook of Alternative Monetary Economics*, pages 35–51. Edward Elgar, Cheltenham, UK.
- Edwards, S. (2024). Inflation and the corruption of currency in Latin America: Chile, 1970–1973. *Journal of Policy History*, 36(4):386–399.
- Espinosa, A. (2021). *The economic impossibility of socialism: an Austrian analysis of Allende’s Chile*. Routledge. GAP FLAG: confirm publisher.
- Espinosa, J. G. and Zimbalist, A. S. (1978). *Economic democracy: workers’ participation in Chilean industry, 1970–1973*. Academic Press, New York.

- Espinosa, V. I. and Véjar, V. (2025). Macroeconomic populism in Chile: Allende and the recession of 1973. *Revista de Historia Económica - Journal of Iberian and Latin American Economic History*, 43(2):309–332.
- Ffrench-Davis, R. (2018). *Reformas económicas en Chile 1973–2017*. Taurus, Santiago.
- Fischer, K. (2009). The influence of neoliberals in Chile before, during, and after Pinochet. *The road from Mont Pèlerin: The making of the neoliberal thought collective*, pages 305–346.
- Foley, D. K. (1982). The value of money, the value of labour-power and the Marxian transformation problem. *Review of Radical Political Economics*, 14(2):37–47.
- Foley, D. K. (2003). Financial fragility in developing economies. *Working Paper Series, Department of Economics, New School University*.
- Frank, A. G. (1967). *Capitalism and underdevelopment in Latin America: historical studies of Chile and Brazil*. Monthly Review Press, New York.
- Garcés, M. (2002). *Tomando su sitio: el movimiento de pobladores de Santiago, 1957–1970*. LOM Ediciones, Santiago.
- Girardi, D. and Bowles, S. (2018). Institution shocks and economic outcomes: Allende’s election, Pinochet’s coup and the Santiago stock market. *Journal of Development Economics*, 134:16–27.
- Gonçalves, S. and Kilian, L. (2004). Bootstrapping autoregressions with conditional heteroskedasticity of unknown form. *Journal of Econometrics*, 123(1):89–120.
- González, F. (2013). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- González, F. and Prem, M. (2026). [title to be confirmed]. GAP FLAG: title, venue, and year unconfirmed — researcher must verify before submission.
- González, F., Prem, M., and Urzúa, F. (2020). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- González, F. and Vial, J. (2021). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Griffith-Jones, S. (1981). *The role of finance in the transition to socialism*. Frances Pinter, London.
- Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing, and inference. *Journal of Econometrics*, 93(2):345–368.
- Harmer, T. (2011). *Allende’s Chile and the Inter-American Cold War*. University of North Carolina Press, Chapel Hill.
- Isch, C., Dörr, T., Fasching, N., Jennings, G., and Watts, D. J. (2026). Quantifying the prevalence and impact of overreaching causal claims in social science. *Nature Human Behaviour*. Advance online publication.
- Ivanova, M. N. (2013). The dollar as world money. *Science & Society*, 77(1):44–71.
- Ivanova, M. N. (2020). Marx’s theory of money: A reappraisal in the light of unconventional monetary policy. *Review of Radical Political Economics*, 52(1):137–151.
- Jaimovich, A. and Toledo-Concha, E. (2021). Failed agrarian reform and the origins of contemporary conflict: evidence from Chile. *The Economic Journal*.
- Jeanne, O. (2007). International reserves in emerging market countries: too much of a good thing? *Brookings papers on Economic activity*, 2007(1):1–79.
- Jenkins, J. A. and Rubin, J., editors (2024). *The Oxford Handbook of Historical Political Economy*. Oxford University Press.
- Jessop, B. (2019). The capitalist state and state power. *The Oxford Handbook of Karl Marx*, pages 298–320.

- Kaldor, N. (1982). *The Scourge of Monetarism*. Oxford University Press, Oxford.
- Kay, C. (1978). Agrarian reform and the class struggle in Chile. In *Chile after 1973*. Latin America Bureau, London. GAP FLAG: confirm booktitle and publisher.
- Kay, C. (2010). *Latin American theories of development and underdevelopment*. Routledge, London.
- King, L. P. and Sznajder, A. (2006). The state-led transition to liberal capitalism: Neoliberal, organizational, world-systems, and social structural explanations of Poland's economic success. *American Journal of Sociology*, 112(3):751–801.
- Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147.
- Kregel, J. (2008). Minsky's cushions of safety, systemic risk and the crisis in the subprime mortgage market. Public Policy Brief 93, Levy Economics Institute of Bard College.
- Kvangraven, I. H. (2021). Beyond the stereotype: Restating the relevance of the dependency research programme. *Development and change*, 52(1):76–112.
- Lakatos, I. (1978). *The Methodology of Scientific Research Programmes: Philosophical Papers Volume 1*. Cambridge University Press, Cambridge.
- Larraín, F. and Meller, P. (1991). The socialist-populist Chilean experience: 1970–1973. In Dornbusch, R. and Edwards, S., editors, *The Macroeconomics of Populism in Latin America*, pages 175–221. University of Chicago Press for NBER, Chicago.
- Larraín, F. and Meller, P. (1991). The socialist-populist Chilean experience, 1970–1973. In Dornbusch, R. and Edwards, S., editors, *The Macroeconomics of Populism in Latin America*, pages 175–221. University of Chicago Press, Chicago.
- Lavoie, M. (2014). *Post-Keynesian Economics: New Foundations*. Edward Elgar Publishing, Cheltenham, UK.
- Leiva, F. I. (2008). *Latin American neostructuralism: the contradictions of post-neoliberal development*. University of Minnesota Press, Minneapolis.
- Loveman, B. (1971). *Antecedentes para el estudio del movimiento campesino chileno: pliegos de peticiones, huelgas y sindicatos agrícolas, 1932–1966*. Instituto de Capacitación e Investigación en Reforma Agraria (ICIRA), Santiago. 2 vols.
- Loveman, B. (1976). *Struggle in the Countryside: Politics and Rural Labor in Chile, 1919–1973*. Indiana University Press, Bloomington, IN.
- MacKinnon, J. G. (1996). Numerical distribution functions for unit root and cointegration tests. *Journal of applied econometrics*, 11(6):601–618.
- Madariaga, A. and Palestini, S., editors (2021). *Dependent capitalisms in contemporary Latin America and Europe*. Palgrave Macmillan.
- Maillet, A., Toro, S., Olivares, A., and Rodríguez, M. I. (2016). Los monjes fuera del monasterio: CIEPLAN y su producción intelectual durante cuatro décadas de política chilena. *Política. Revista de Ciencia Política*, 54(1):189–218.
- Marini, R. M. (1973). *Dialéctica de la dependencia*. Era, México.
- Meller, P. (2000). *The Unidad Popular and the Pinochet dictatorship: a political analysis*. Macmillan, Basingstoke. GAP FLAG: confirm exact title — may be chapter in edited volume or different title.
- Milos, P. (2008). *Frente Popular en Chile: su configuración, 1935–1938*. LOM Ediciones, Santiago.
- Minsky, H. P. (1986). *Stabilizing an Unstable Economy*. Yale University Press, New Haven.
- Moore, B. J. (1988). *Horizontalists and Verticalists: The Macroeconomics of Credit Money*. Cambridge University Press, Cambridge.

- Morris, J. O. and Oyaneder, R. (1962). *Afiliación y finanzas sindicales en Chile, 1932–1959*. INSORA, Universidad de Chile, Santiago.
- Murphy, E. (2015). *For a Proper Home: Housing Rights in the Margins of Urban Chile, 1960–2010*. University of Pittsburgh Press, Pittsburgh, PA.
- Nievas, G. and Piketty, T. (2025). The world balance of payments database: 1900–2024. *World Inequality Lab Working Paper*, 2025/04.
- Osorio, S. and Polanco, D. (2021). Tasas de afiliación sindical y tamaño promedio de los sindicatos en Chile, 1932–2010. Minuta 2, Repositorio de Estadísticas Sindicales (RES).
- Palley, T. I. (2013). Horizontalists, verticalists, and structuralists: The theory of endogenous money. In *Financialization: The Macroeconomics of Finance Capital*, pages 109–126. Palgrave Macmillan, London.
- Palma, G. (1984). From an export-led to an import-substituting economy: Chile 1914–1939. In Thorp, R., editor, *Latin America in the 1930s: The Role of the Periphery in World Crisis*, pages 50–80. Palgrave Macmillan, London.
- Palma, J. G. (2016). The “dependency school” and its aftermath: why latin america’s critical thinking switched from one type of absolute certainties to another’. *Handbook of alternative theories of economic development*, pages 386–415.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition.
- Petras, J. F. and Morley, M. H. (1975). *The United States and Chile: imperialism and the overthrow of the Allende government*. Monthly Review Press, New York.
- Pinto Santa Cruz, A. (1958). *Chile, un caso de desarrollo frustrado*. Editorial Universitaria, Santiago.
- Ramos, J. (1980). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Ramos, S. (1979). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- República de Chile (1947). Ley 8.811: Agrega disposiciones al Código del Trabajo, relativas a la organización sindical de los obreros agrícolas. Diario Oficial, 29 de julio de 1947. Texto original; promulgada el 8 de julio de 1947.
- República de Chile (1948). Ley 8.987: Modifica la ley 6.026 sobre seguridad interior del estado, en la forma que indica. Diario Oficial, 3 de septiembre de 1948. Texto original de la Ley de Defensa Permanente de la Democracia.
- República de Chile (1967). Ley 16.625: Sindicación campesina. Diario Oficial, 29 de abril de 1967. Texto original; promulgada el 26 de abril de 1967.
- Rochon, L.-P. (1999). *Credit, Money, and Production: An Alternative Post-Keynesian Approach*. Edward Elgar Publishing, Cheltenham, UK.
- Rochon, L.-P. and Rossi, S. (2013). Endogenous money: the evolutionary versus revolutionary views. *Review of Keynesian Economics*, 1(2):210–229.
- Rodríguez-Asensio, D. and López-García, A. M. (2026). Transmission of macroeconomic shocks in housing markets: A threshold VAR approach. *Journal of Real Estate Finance and Economics*.
- Rodríguez Weber, J. E. (2018). Alta desigualdad en América Latina: desde cuándo y por qué. *Documentos de Trabajo en Historia Económica*.
- Roncaglia, A. (2006). Why should economists study the history of economic thought? *The European Journal of the History of Economic Thought*, 13(1):1–18.
- Ruiz Bruzzone, C. (2021). Financialization and peripheral business power in the Chilean pension system. In Madariaga, A. and Palestini, S., editors, *Dependent Capitalisms in Contemporary*

- Latin America and Europe*, pages 231–257. Palgrave Macmillan.
- Runge, J., Nowack, P., Kretschmer, M., Flaxman, S., and Sejdinovic, D. (2019). Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, 5(11):eaau4996.
- Schlotterbeck, M. E. (2018). *Beyond the Vanguard: Everyday Revolutionaries in Allende’s Chile*. University of California Press.
- Sideri, S., editor (1979). *Chile 1970–73: economic development and its international setting: self-criticism of the Unidad Popular government’s policies*. Martinus Nijhoff, The Hague.
- Silva, E. (2007). The import-substitution model: Chile in comparative perspective. *Latin American Perspectives*, 34(3):67–90.
- Silva, P. (1991). Technocrats and politics in Chile: from the Chicago Boys to the CIEPLAN Monks. *Journal of Latin American Studies*, 23(2):385–410.
- Sims, C. A. and Zha, T. (1999). Error bands for impulse responses. *Econometrica*, 67(5):1113–1155.
- Sklair, L. (2000). The transnational capitalist class and the discourse of globalisation. *Cambridge Review of International Affairs*, 14(1):67–85.
- Sossdorf Gonzalez, F. (2024). [trade paper title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Stallings, B. (1978). *Class conflict and economic development in Chile, 1958–1973*. Stanford University Press, Stanford.
- Stallings, B. and Zimbalist, A. (1975). The political economy of Chilean underdevelopment. *Latin American Perspectives*, 2(4):69–88. GAP FLAG: confirm volume, number, pages.
- Sweezy, P. M. and Magdoff, H. (1974). *Revolution and counter-revolution in Chile*. Monthly Review Press, New York.
- Thielemann, L. (2018). La rudeza pagana: sobre la radicalización del movimiento obrero en los largos sesenta. Chile, 1957–1970. *Izquierdas*, 44:114–133.
- Thielemann Hernández, L. (2023). *1957. El proletariado invade Santiago. Un año de revuelta, tomas y ruptura estratégica en el alba de la metrópolis chilena*. Ariadna Ediciones and Tesis XII Editores, Santiago de Chile.
- Valdés, J. G. (1995). *Pinochet’s economists: the Chicago school of economics in Chile*. Cambridge University Press, Cambridge.
- Vasudevan, R. (2009). From the gold standard to the floating dollar standard: An appraisal in the light of Marx’s theory of money. *Review of Radical Political Economics*, 41(4):473–491.
- Vasudevan, R. (2016). Financialization, distribution and accumulation: a circuit of capital model with a managerial class. *Metroeconomica*, 67(2):397–428.
- Vasudevan, R. (2019). Money, power, and capitalism: Marx’s theory of money and the contemporary state-credit standard. In Gupta, S., Basu, M. M., and Babones, S., editors, *Karl Marx’s Life, Ideas, and Influences: A Critical Examination on the Bicentenary*, pages 229–246. Palgrave Macmillan, Cham.
- Vernengo, M. (2021). The consolidation of dollar hegemony after the collapse of Bretton Woods: Bringing power back in. *Review of Political Economy*, 33(4):529–551.
- Vidal, M. (2019). Geriatric capitalism: Stagnation and crisis in western capitalism. In *The Oxford Handbook of Karl Marx*, pages 581–606. Oxford University Press.
- Weisskopf, T. E. (1979). Marxian crisis theory and the rate of profit in the postwar u.s. economy. *Cambridge Journal of Economics*, 3(4):341–378.
- Winn, P. (1986). *Weavers of revolution: the Yarur workers and Chile’s road to socialism*. Oxford University Press, New York.

- Winn, P. (2016). *La revolución chilena*. LOM Ediciones, Santiago. 2016 catalogue version; edition number not verified.
- Zavaleta Mercado, R. (1974). El poder dual en América Latina: estudio de los casos de Bolivia y Chile. *Cuadernos Políticos*.
- Zoeller, C. J. P. (2019). Closing the gold window: The end of Bretton Woods as a contingency plan. *Politics & Society*, 47(1):3–22.

Appendix A Formalization of the Central Bank Solvency Ratio and Balance-of-Payments Accounting

A.1 This appendix develops the formal balance-sheet accounting, the exact discrete law of motion, and the structural constraint-binding conditions governing the central bank **Solvency Ratio** (SolvR_t). Every accounting step is an exact discrete identity; no linear approximation or continuous-time distortion enters the derivation.

A.1 The Solvency Ratio in Nominal Terms

A.2 All variables are denominated in current domestic currency (Chilean pesos/escudos, CLP) at time t , with the nominal exchange rate convention defined as e_t (CLP per USD), such that a nominal depreciation corresponds to $e_t \uparrow$.

A.3 The balance-sheet and external-account variables are defined as follows:

- M_t : domestic monetary obligations (CLP), representing the liquid liabilities of the consolidated domestic financial system underwritten by the central bank.
- IR_t : stock of gross international reserves (USD), representing claims denominated in world money.
- e_t : nominal exchange rate (CLP per USD).
- $F_t \equiv e_t \cdot IR_t$: stock of world money held by the central bank, valued in domestic currency (CLP).
- $\text{BoP}_t \equiv \Delta F_t$: net balance-of-payments flow (CLP).
- $g_{M,t} \equiv \Delta M_t / M_t$: simple growth rate of domestic money obligations.
- $g_{F,t} \equiv \Delta F_t / F_t = \text{BoP}_t / (e_t \cdot IR_t)$: simple growth rate of the world-money asset base.
- X_t : export volume of primary and manufactured goods.
- Q_t : import volume of intermediate, consumption, and capital goods.
- $P_{Q,t}^{USD}$: import price index in foreign currency (USD).
- $\text{ToT}_t \equiv P_{X,t}^{USD} / P_{Q,t}^{USD}$: terms of trade.
- D_t^{ext} : stock of external debt (USD).
- r_t : average nominal interest rate on external debt.
- FDI_t : net foreign direct investment inflows (USD).
- Π_t : net foreign profit and dividend remittances (USD).
- T_t : net current external transfers (CLP).

Import volumes carry the symbol Q_t throughout to prevent notation collisions with domestic money M_t . Discrete growth rates are evaluated on the contemporaneous base to maintain algebraic exactness.

A.4 I define the central bank Solvency Ratio (SolvR_t) in nominal terms as:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} = \frac{F_t}{M_t} \quad (9)$$

The Solvency Ratio measures the stock of world-money assets held by the central bank per unit of unanchored domestic credit obligations underwritten by the monetary authority. Its algebraic inverse, $\text{LevR}_t \equiv 1/\text{SolvR}_t = M_t/(e_t \cdot IR_t)$, expresses the corresponding balance-sheet leverage ratio. I center the analytical framework on the Solvency Ratio because it directly formalizes central bank reserve backing and operationalizes the transposition of Minsky's financial postures to the peripheral central bank's own balance sheet (Foley, 2003; Chamorro-Futinico, 2025b) (Section 3.5).

A.5 Taking continuous growth rates of Eq. (9) yields the continuous growth decomposition:

$$g_{\text{SolvR},t} = g_{F,t} - g_{M,t} = g_{e,t} + g_{IR,t} - g_{M,t} \quad (10)$$

Equation (10) isolates three distinct structural channels driving peripheral balance-sheet solvency:

1. **External Balance-of-Payments Channel** ($g_{IR,t}$, expanding solvency): Export revenues and net foreign capital inflows accumulate foreign exchange reserves, bolstering central bank solvency. Current account deficits, foreign debt service, and transnational profit repatriation drain reserves, pulling SolvR_t downward.
2. **Valuation Channel** ($g_{e,t}$, expanding nominal solvency): Nominal currency depreciation ($e_t \uparrow$) increases the domestic-currency accounting valuation of foreign assets ($F_t = e_t \cdot IR_t$). While this mechanically cushions SolvR_t in domestic currency units, it carries acute real macroeconomic costs by inflating imported intermediate input costs and triggering domestic wage-price defense spirals.
3. **Domestic Dilution Channel** ($g_{M,t}$, compressing solvency): Broad money creation, fiscal monetization, and emergency Lender-of-Last-Resort liquidity expand domestic obligations from below, continuously diluting the world-money backing of the monetary base.

A.2 Current and Capital Account Decomposition

A.6 The evolution of the central bank's foreign-currency asset base is governed by the external balance of payments. In domestic currency terms, the current account (CA_t) and capital account (KA_t) are specified as:

$$CA_t = e_t \cdot P_{Q,t}^{USD}(\text{ToT}_t \cdot X_t - Q_t) - e_t \cdot r_t \cdot D_t^{\text{ext}} + T_t \quad (11)$$

$$KA_t = e_t \cdot \Delta D_t^{\text{ext}} + e_t \cdot FDI_t - e_t \cdot \Pi_t \quad (12)$$

The sum of the current and capital accounts determines the net balance-of-payments flow (BoP_t), which dictates the net change in world-money holdings:

$$\Delta F_t \equiv \Delta(e_t \cdot IR_t) = CA_t + KA_t \equiv \text{BoP}_t \quad (13)$$

A.3 Derivation of the Discrete Solvency Law of Motion

A.7 From Eq. (9), the total stock of world-money assets held by the central bank decomposes into the product of the Solvency Ratio and domestic monetary obligations:

$$F_t = \text{SolvR}_t \cdot M_t \quad (14)$$

Applying the exact discrete product rule $\Delta(A_t B_t) = B_t \Delta A_t + A_{t-1} \Delta B_t$ with $A_t = \text{SolvR}_t$ and $B_t = M_t$ yields:

$$\Delta F_t = M_t \cdot \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \Delta M_t \quad (15)$$

Dividing Eq. (15) through by M_t and substituting $\Delta F_t = \text{BoP}_t$ from Eq. (13):

$$\frac{\text{BoP}_t}{M_t} = \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \frac{\Delta M_t}{M_t} \quad (16)$$

Rearranging terms isolates the exact discrete law of motion for the Solvency Ratio:

$$\boxed{\Delta \text{SolvR}_t = \frac{\text{BoP}_t}{M_t} - \text{SolvR}_{t-1} \cdot g_{M,t}} \quad (17)$$

where $g_{M,t} \equiv \Delta M_t / M_t$ is the simple growth rate of domestic monetary liabilities.

A.8 Equation (17) delivers an intuitive and powerful economic decomposition:

- The first term, BoP_t / M_t , represents the **external liquidity injection rate**—net foreign exchange inflows scaled directly by the existing domestic credit pyramid. Balance-of-payments surpluses expand the asset base and bolster solvency.
- The second term, $\text{SolvR}_{t-1} \cdot g_{M,t}$, represents the **domestic dilution drag**. Rapid domestic money growth ($g_{M,t} > 0$) dilutes the inherited backing ratio. The presence of the lagged state variable SolvR_{t-1} is an exact algebraic consequence of the discrete product rule: domestic monetary expansion dilutes the preexisting stock of world-money cover.

A.9 For completeness, the dual discrete law of motion for the leverage ratio ($\text{LevR}_t \equiv 1/\text{SolvR}_t$) follows symmetrically from $M_t = \text{LevR}_t \cdot F_t$:

$$\Delta \text{LevR}_t = \frac{\Delta M_t}{e_t \cdot IR_t} - \text{LevR}_{t-1} \cdot g_{F,t} \quad (18)$$

Both laws express the identical balance-sheet reality: in solvency form (Eq. 17), foreign inflows drive recovery while domestic monetization acts as a dilution drag; in leverage form (Eq. 18), domestic money creation drives leverage expansion while reserve accumulation acts as an absorption multiplier.

A.4 Analytical Convexity and Non-Linear Reserve Exhaustion

A.10 The structural vulnerability of peripheral central banks is amplified by the non-linear curvature of balance-sheet solvency. Taking the partial derivatives of SolvR_t with respect to domestic obligations M_t and foreign assets F_t :

$$\frac{\partial \text{SolvR}_t}{\partial M_t} = -\frac{F_t}{M_t^2} < 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial M_t^2} = \frac{2F_t}{M_t^3} > 0 \quad (\forall M_t > 0, F_t > 0) \quad (19)$$

$$\frac{\partial \text{SolvR}_t}{\partial F_t} = \frac{1}{M_t} > 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial F_t^2} = 0 \quad (20)$$

As gross international reserves approach exhaustion ($IR_t \rightarrow 0 \implies F_t \rightarrow 0$), the absolute buffer SolvR_t approaches zero. In this depleted state, any incremental withdrawal of foreign currency or external import shock triggers an acute percentage collapse in central bank credibility, permanently crippling the authority's capacity to anchor expectations.

A.5 Constraint-Binding Conditions and Structural Minskian Regimes

A.11 The central bank’s solvency position deteriorates ($\Delta \text{SolvR}_t < 0$) whenever domestic monetary dilution outpaces net balance-of-payments inflows:

$$\Delta \text{SolvR}_t < 0 \iff \frac{\text{BoP}_t}{M_t} < \text{SolvR}_{t-1} \cdot g_{M,t} \iff g_{M,t} > \frac{\text{BoP}_t}{F_{t-1}} \quad (21)$$

Even during an external commodity price boom ($\text{BoP}_t > 0$, such as the 1971–1972 copper price high), central bank solvency deteriorates if domestic credit expansion outpaces the growth rate of foreign reserves.

A.12 In my structural Post-Keynesian framework, monetization is an endogenous institutional response to distributive conflict and foreign-exchange bottlenecks rather than an exogenous policy choice. The state-dependent parameter $\beta_{gM}(\text{SolvR}_{t-1})$ characterizes the regime-conditional absorption capacity of the peripheral monetary system. Following the 3-regime Threshold Vector Autoregression estimated in Section 5, the macro-system transitions across three discrete Minskian states:

1. **Hedge Regime** ($\text{SolvR}_{t-1} > \hat{\gamma}_H = 0.21$): Reserve backing comfortably validates domestic liquid obligations. The central bank credibly maintains currency stability, and domestic monetary expansion is absorbed without price instability ($\beta_{\text{Hedge}} \approx 0.00\%$).
2. **Speculative Regime** ($0.14 < \text{SolvR}_{t-1} \leq 0.21$): Reserve buffers are partially depleted. Domestic credit growth begins to strain central bank credibility, and the pass-through of money to prices increases to $+0.54\%$.
3. **Forced Ponzi Regime** ($\text{SolvR}_{t-1} \leq \hat{\gamma}_P = 0.14$): Foreign exchange reserves are exhausted. The central bank is forced into emergency Lender-of-Last-Resort monetization to avert systemic enterprise insolvencies. Because world-money backing is gone, emergency liquidity cannot be absorbed by the balance sheet and passes directly into hyperinflation and currency flight ($\beta_{\text{Ponzi}} = +4.54\%$).

This threshold formalization establishes that hyperinflation is a non-linear property of balance-sheet solvency exhaustion under external financial subordination, not a linear artifact of money creation.

Appendix B Atlas of High-Frequency Monthly TVAR Impulse Responses Across Solvency Regimes (1960–1980)

C.1 To evaluate the non-linear macroeconomic transmission mechanism under balance-sheet exhaustion without temporal aggregation distortions, this appendix presents the comprehensive Generalized Impulse Response Function (GIRF) atlas across sixty dynamic trajectories. The empirical system comprises the four-variable endogenous monthly vector ($N = 252$, 1960:01–1980:12):

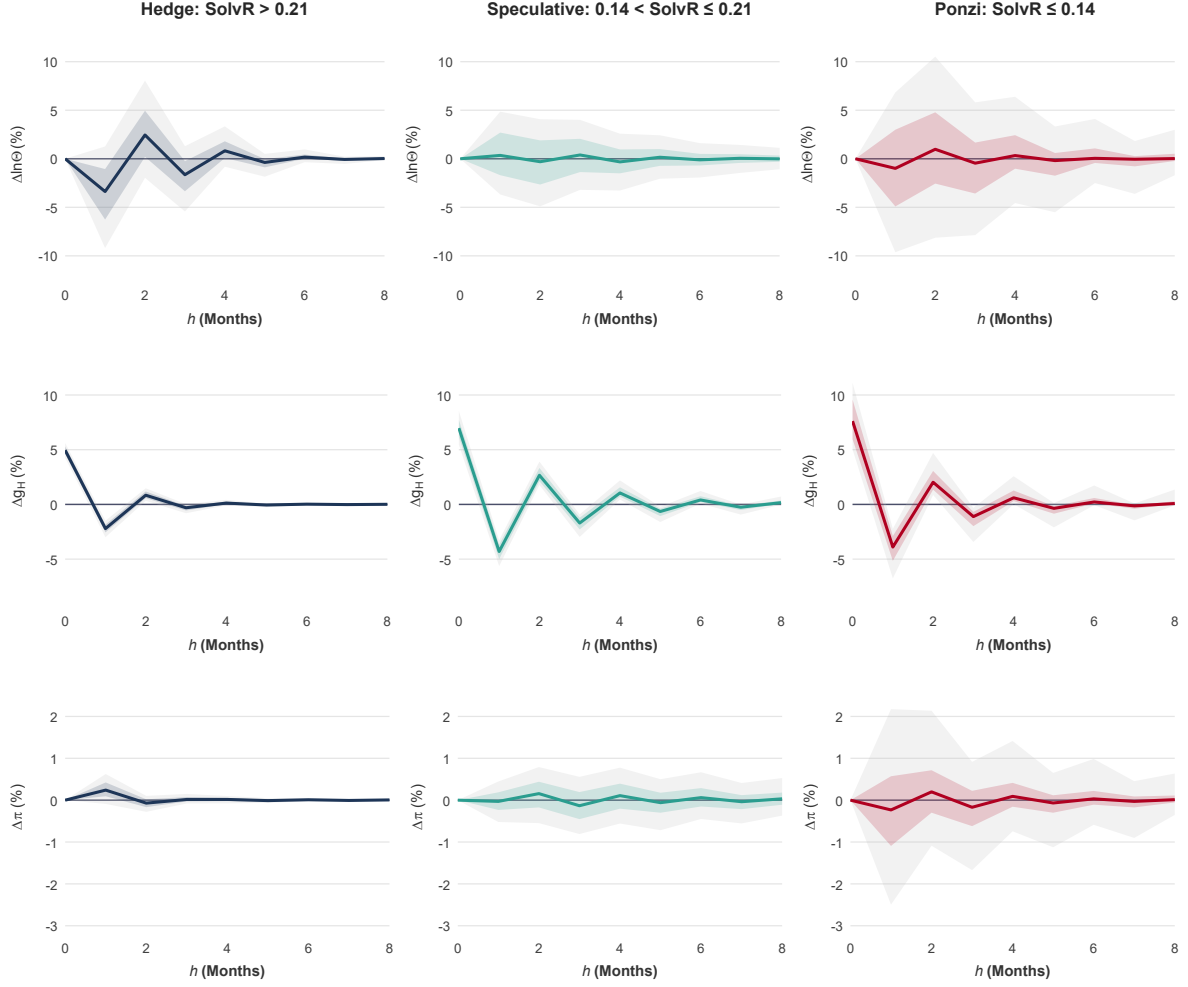
$$y_t = [g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]' \quad (22)$$

conditioned on lagged Central Bank reserve solvency $\text{SolvR}_{t-1} \equiv \frac{e_{t-1} \cdot IR_{t-1}}{M1_{t-1}}$ across the three Minskian solvency regimes: Hedge ($\text{SolvR} > 0.21$), Speculative ($0.14 < \text{SolvR} \leq 0.21$), and Forced Ponzi ($\text{SolvR} \leq 0.14$).

C.2 All impulse responses are computed via the nested two-loop simulation algorithm synthesized from Koop et al. (1996), Balke (2000), and Afonso et al. (2018):

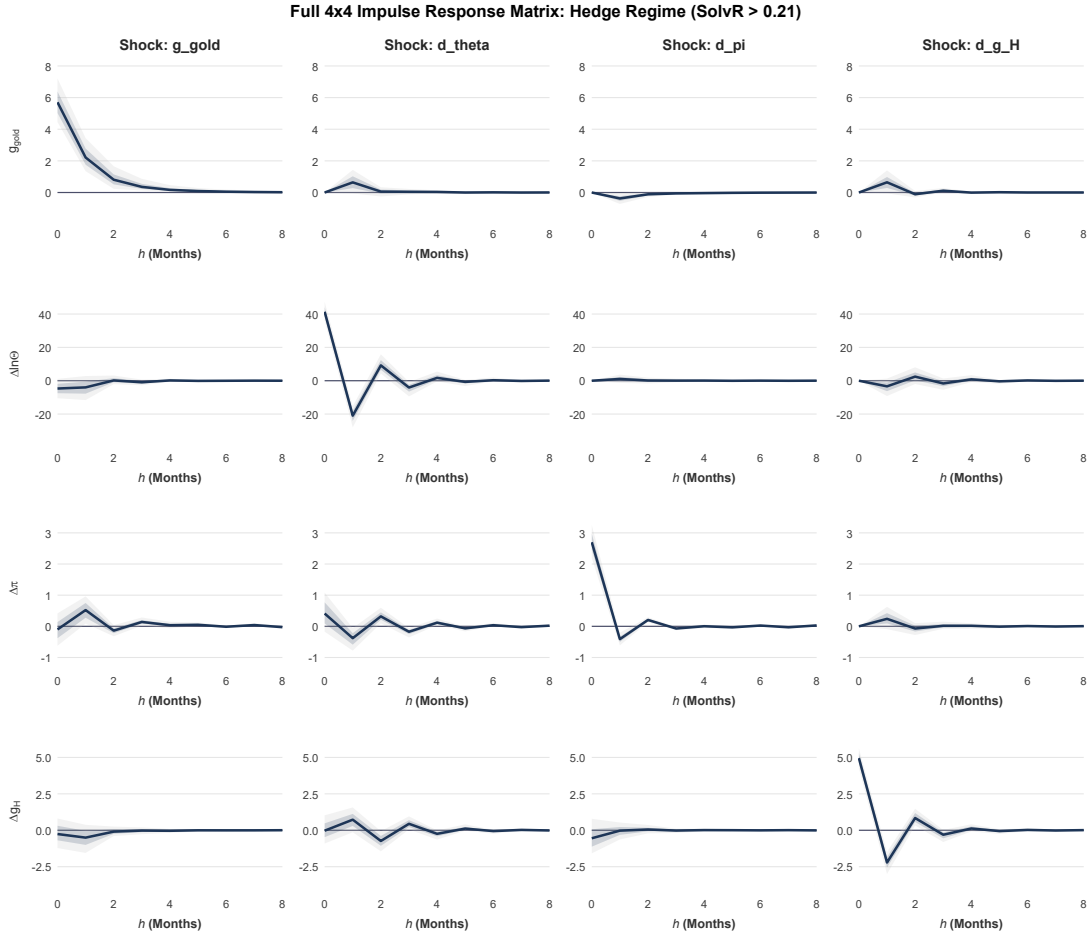
1. **Parameter Uncertainty (Outer Loop):** $B = 500$ stratified residual bootstrap replications are executed, drawing residuals independently from each regime’s empirical residual covariance matrix $\hat{\Sigma}_k$ to preserve regime-specific conditional heteroskedasticity ([Gonçalves and Kilian, 2004](#)).
2. **History-Conditioned Shock Propagation (Inner Loop):** For each historical starting month $t_0 \in \mathcal{T}_k$, orthogonalized Cholesky structural innovations $v_0 = P_k e_j$ are propagated forward over $h = 0, \dots, 8$ months along the empirical regime sequence.
3. **Credibility Ribbons:** Following [Sims and Zha \(1999\)](#) and [Rodríguez-Asensio and López-García \(2026\)](#), each panel displays the empirical median trajectory (solid line), the primary 68% confidence ribbon (16th–84th percentiles, darker shading), and the 95% conservative bounds (2.5th–97.5th percentiles, lighter shading), evaluated against the neutral grey horizontal zero reference line.

Figure 3: Master Nonlinear Generalized Impulse Response Matrix: Real Structural Imbalance, Base Money, and Inflation Across Minskian Solvency Regimes



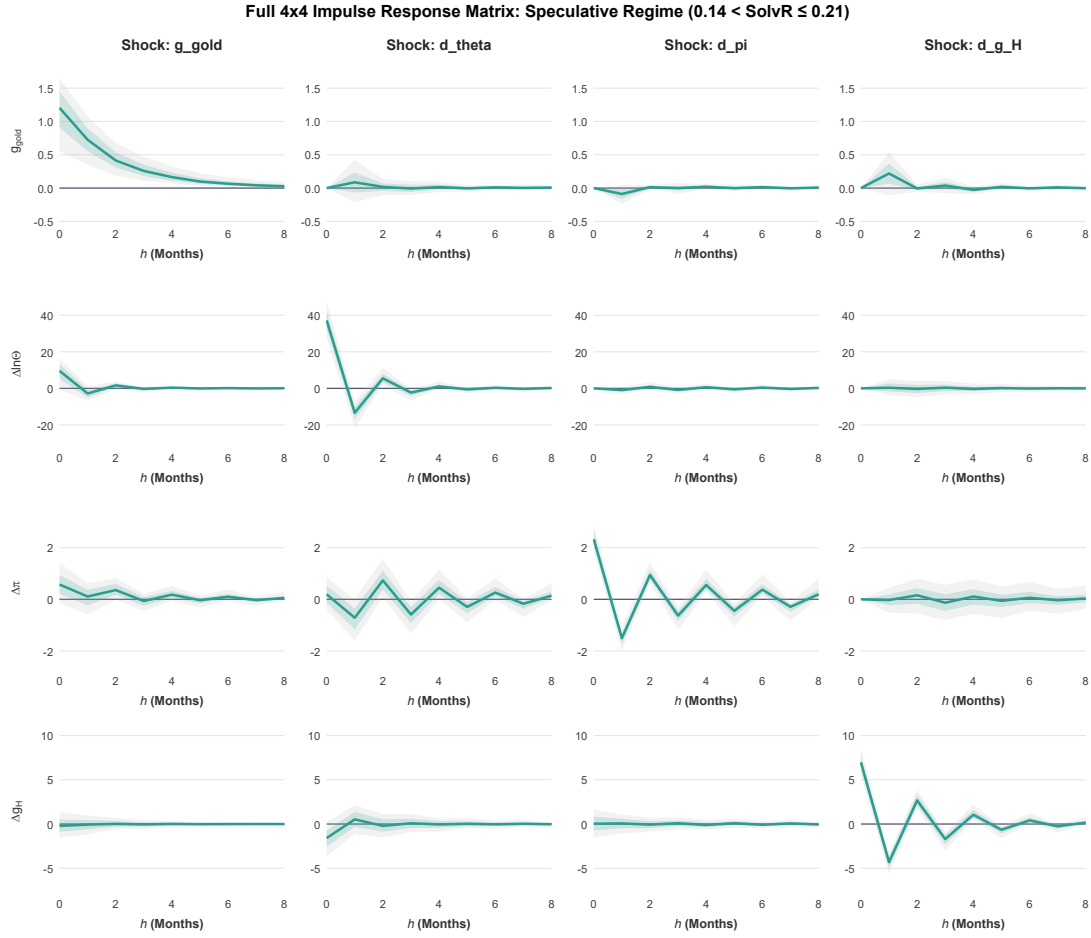
Notes: Generalized Impulse Response Functions (GIRFs) of the Real Structural Imbalance ($\Delta \ln \Theta_t$), base money growth ($\Delta g_{H,t}$), and consumer price inflation ($\Delta \pi_t$) to a $+1\sigma$ orthogonalized base money expansion shock (Δg_H), estimated from a 3-regime TVAR(1) with predetermined delay $d = 1$ (SolvR_{t-1}). Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified non-parametrically by the PCMCi contemporaneous DAG (Appendix E). Solid colored lines display empirical median responses across Minskian postures (navy for Hedge, teal for Speculative, crimson for Forced Ponzi); darker and lighter shaded polygons display 68% and 95% stratified residual bootstrap credibility bands ($B = 500$ replications) following Sims and Zha (1999) and Rodríguez-Asensio and López-García (2026), evaluated against the neutral grey horizontal zero reference line over horizon $h = 8$ months. Responses are expressed in percentage-point monthly deviations. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12, $N = 250$). Complete 4×4 regime response matrices covering all 48 shock-response combinations are documented in Figures 4–8.

Figure 4: Full 4×4 Impulse Response Matrix: Hedge Regime ($\text{SolvR} > 0.21$)



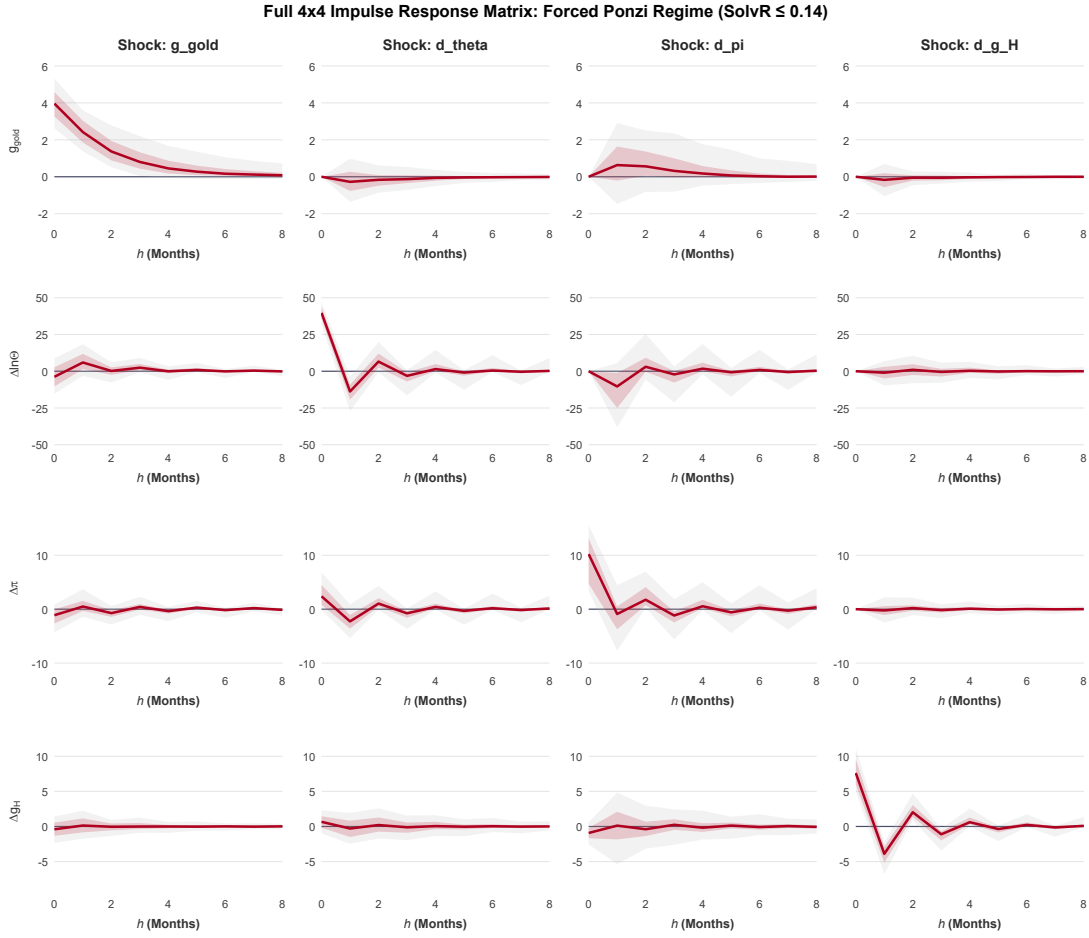
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Hedge regime ($\text{SolvR}_{t-1} > 0.21$, $N = 143$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous response variables ($g_{P,\text{gold}}, \Delta \ln \Theta, \Delta \pi, \Delta g_H$). Identification follows the Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ authorized by the PCMCi DAG. Trajectories trace horizons $h = 0 \dots 8$ months; shaded bands represent 68% (darker) and 95% (lighter) stratified residual bootstrap credibility bands ($B = 500$). Responses are measured in percentage-point monthly deviations. Under anchored external solvency, base money shocks (Column 4) produce statistically zero price acceleration. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 5: Full 4×4 Impulse Response Matrix: Speculative Regime ($0.14 < \text{SolvR} \leq 0.21$)



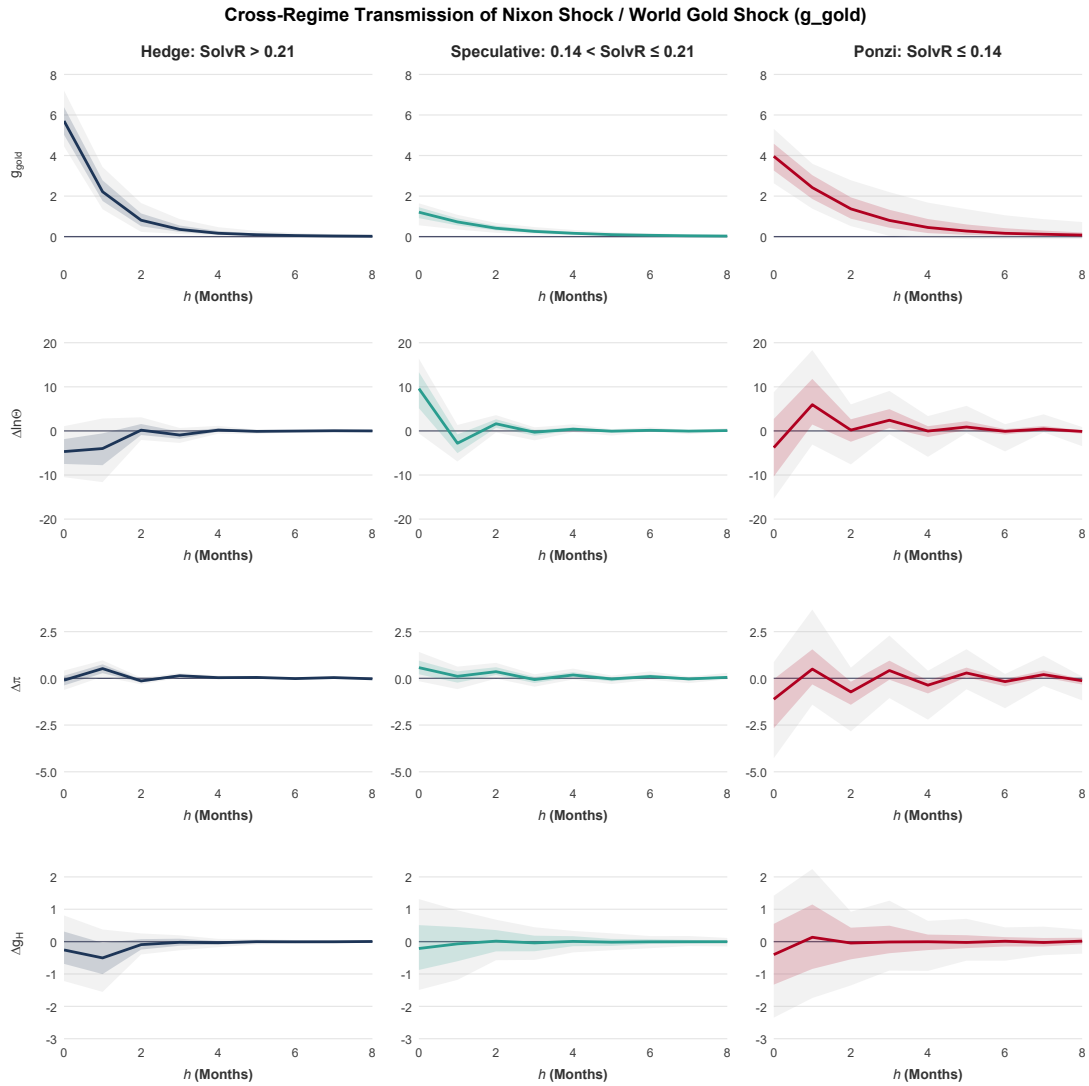
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Speculative regime ($0.14 < \text{SolvR}_{t-1} \leq 0.21$, $N = 62$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous responses. Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$ replications) over horizon $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. As foreign reserve coverage erodes, external gold price shocks (Column 1) pass violently into domestic inflation. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 6: Full 4×4 Impulse Response Matrix: Forced Ponzi Regime ($\text{SolvR} \leq 0.14$)



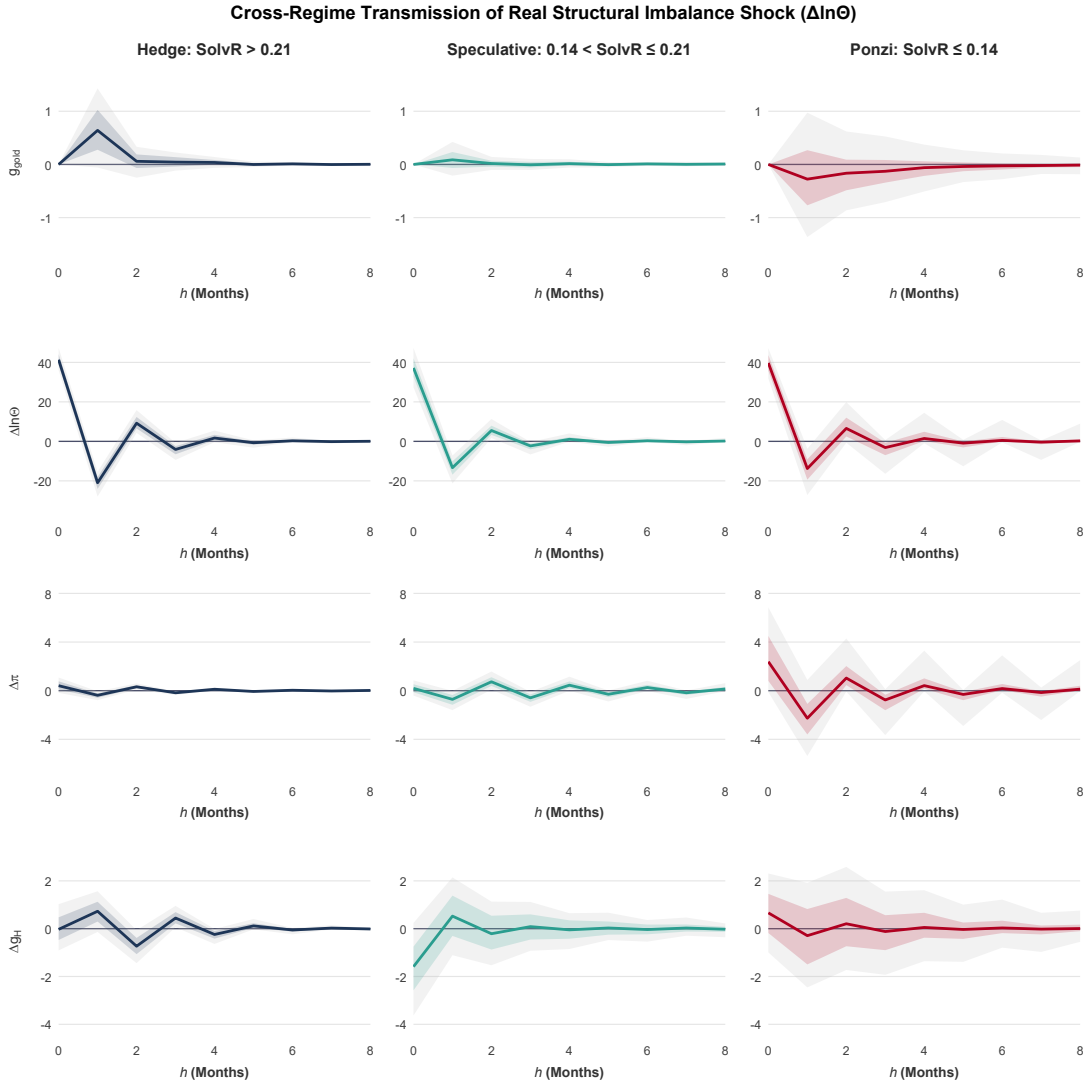
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Forced Ponzi regime ($\text{SolvR}_{t-1} \leq 0.14$, $N = 45$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous responses. Structural identification follows the recursive Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$) across $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. Under liquid reserve exhaustion, domestic inflation shocks (Column 3) trigger immediate, persistent defensive base money accommodation (Row 4), while monetary shocks (Column 4) unleash explosive price acceleration. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 7: Cross-Regime Transmission Atlas: Nixon Shock and World Gold Price Growth ($g_{P,\text{gold}}$)



Notes: Comparative cross-regime Generalized Impulse Response Functions (GIRF) to a $+1\sigma$ international gold price growth shock ($g_{P,\text{gold}}$) across Hedge ($\text{SolvR} > 0.21$, solid navy line), Speculative ($0.14 < \text{SolvR} \leq 0.21$, dashed gold line), and Forced Ponzi ($\text{SolvR} \leq 0.14$, dotted crimson line) regimes from the 3-regime TVAR(1) ($d = 1$). Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCI DAG. Shaded areas represent 68% and 95% stratified residual bootstrap bands ($B = 500$) across horizon $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are expressed in percentage-point monthly deviations. World-money stagflation transmits into severe domestic cost wedges only when the Central Bank lacks foreign reserve cushions. Data sources: London Bullion Market, Central Bank of Chile, and INE (1960:01–1980:12).

Figure 8: Cross-Regime Transmission Atlas: Real Structural Imbalance Shock ($\Delta \ln \Theta$)



Notes: Comparative cross-regime Generalized Impulse Response Functions (GIRF) to a $+1\sigma$ Real Structural Imbalance shock ($\Delta \ln \Theta$) across Hedge (SolvR > 0.21, solid navy line), Speculative ($0.14 < \text{SolvR} \leq 0.21$, dashed gold line), and Forced Ponzi ($\text{SolvR} \leq 0.14$, dotted crimson line) regimes from the 3-regime TVAR(1) ($d = 1$). Structural identification follows the recursive Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$) across $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. Import asphyxiation triggers immediate output contraction and explosive inflation jumps under balance-sheet insolvency. Data sources: Central Bank of Chile (BCCh) and INE (1960:01–1980:12).

Appendix C Integration Order Diagnostics and Complete Unit Root Battery (Chile, 1920–2010)

D.1 This appendix documents the econometric unit-root analysis evaluating the stochastic properties and integration order of 19 candidate macroeconomic and balance-of-payments variables for Chile over the secular horizon (1920–2010, $T = 91$). Using the `urca` package in R, I implement a comprehensive four-test battery:

1. **Augmented Dickey-Fuller (ADF)** test with drift (`ur.df`, 5% critical value: -2.89);

2. **Elliott-Rothenberg-Stock (DF-GLS)** test with constant (`ur.ers`, 5% critical value: -1.94);
3. **Kwiatkowski-Phillips-Schmidt-Shin (KPSS)** test for level stationarity around a constant (`ur.kpss`, 5% critical value: 0.463);
4. **Zivot-Andrews (ZA)** test allowing for a single endogenous structural break in both intercept and trend (`ur.za`, 5% critical value: -5.08).

D.2 Table 4 reports the empirical test statistics, endogenous break dates, and formal stationarity classifications. The candidate series fall into two distinct operational categories:

- **$I(0)$ Pure Stationary:** Series that reject the unit-root null in standard ADF/DF-GLS tests and fail to reject the stationary null in KPSS tests without requiring structural break corrections (e.g., base money growth g_{M0} , narrow money growth g_{M1} , CPI inflation π_t , USD reserve growth g_{IR} , and fiscal deficit as % of GDP).
- **$I(0)$ Conditional on an Endogenous Structural Break:** Series where secular institutional realignments (such as the 1934 post-Depression agricultural collapse, the 1944 wartime shift, the 1958 Klein-Saks stabilization, or the 1981 debt crisis) generate spurious unit-root non-rejection under standard linear tests, but decisively reject the unit-root null once endogenous break dynamics are accommodated.

Table 4: Unit Root Test Battery and Stationarity Diagnostics for Chilean Candidate System Variables (1920–2010)

Variable / Measure	N	ADF Stat	ERS Stat	KPSS Stat	ZA Stat	ZA Break	Stationarity Classification
Growth of Monetary Base ($g_{M0,t}$)	90	-2.442	-2.874	0.317	-5.451	1977	I(0) Pure Stationary
Growth of Narrow Money ($g_{M1,t}$)	90	-2.511	-2.401	0.386	-4.623	1969	I(0) Pure Stationary
Growth of Broad Money ($g_{M2,t}$)	90	-2.325	-2.147	0.415	-4.369	1969	I(0) Pure Stationary
CPI Inflation Rate (π_t)	90	-3.403	-2.416	0.277	-4.103	1971	I(0) Pure Stationary
Reserve Growth in USD ($g_{IR,USD,t}$)	84	-5.639	-4.083	0.080	-6.958	1976	I(0) Pure Stationary
Reserve Growth in CLP ($g_{IR,CLP,t}$)	84	-9.297	-4.016	0.054	-7.799	1965	I(0) Pure Stationary
Nominal Exchange Rate Growth ($g_{e,t}$)	90	-4.071	-2.833	0.218	-5.487	1971	I(0) Pure Stationary
Copper Price Growth Rate ($g_{\text{copper},t}$)	90	-7.768	-1.662	0.149	-8.479	2003	I(0) Pure Stationary
Fiscal Deficit (% of GDP)	91	-4.742	-2.565	0.213	-5.181	1974	I(0) Pure Stationary
Copper Tax Revenue Share (%)	86	-4.266	-1.843	0.123	-4.354	1970	I(0) Pure Stationary
Astorga Wage Share (W/Y)	91	-3.288	-1.469	1.492	-6.506	1958	I(0) Conditional Break (1958)
Model 3 Detrended Wage Share Residuals	91	-5.161	-2.036	0.166	-6.435	1958	I(0) Pure Stationary
Real Agri / Manuf Labor Productivity Ratio	91	-1.503	-1.402	0.749	-5.343	1934	I(0) Conditional Break (1934)
WBOP Total Trade Balance (% of GDP)	91	-3.555	-0.810	0.537	-6.721	1981	I(0) Conditional Break (1981)
WBOP Primary Income Outflow Drain (% of GDP)	91	-2.105	-1.484	0.380	-3.084	1929	I(0) Pure Stationary
External NFA Liabilities / Assets Leverage Ratio	91	-0.653	-0.822	0.811	-4.749	1944	I(0) Conditional Break (1944)
Clio-Lab Terms of Trade Index (1920–2010)	91	-2.029	-2.302	0.259	-3.090	1926	I(0) Pure Stationary
Structuralist Import Capacity Ratio ($\text{TOT} \times X / M$)	91	-2.237	-1.425	0.473	-4.112	1995	I(0) Conditional Break (1995)
Model 3 Detrended Import Capacity Ratio Residuals	91	-2.025	-1.614	0.271	-4.123	1995	I(0) Pure Stationary

Notes: Primary sample: 1920–2010 annual dataset ($T = 91$). ADF Stat reports `ur.df` with drift (5% CV: -2.89). ERS Stat reports DF-GLS with constant (5% CV: -1.94). KPSS Stat reports level stationarity around a constant (5% CV: 0.463). ZA Stat reports Zivot-Andrews minimum t -statistic evaluating endogenous structural breaks in intercept and trend (5% CV: -5.08). The stationarity classification confirms that all series ingested into the annual and monthly VAR systems are either $I(0)$ pure stationary or $I(0)$ conditional on documented institutional structural breaks.

D.3 These unit-root results confirm that the empirical specifications deployed in Section 4.4 and Appendices B and C satisfy time-series stationarity requirements, protecting the econometric tournament against spurious regression artifacts.

Appendix D Archival Data Provenance, Variable Dictionary, and Splicing Protocols

E.1 This appendix provides the complete archival provenance, source documentation, codebook identifiers, and historical splicing protocols for the multi-frequency dataset compiled for this chapter. The empirical design coordinates two distinct frequency tiers:

1. **Monthly High-Frequency Panel (1960:01–1980:12, $N = 252$ months):** Compiled from the Central Bank of Chile REST API (`SieteRestWS`) for monetary aggregates, exchange rates, and official consumer prices, the physical production and customs trade volume series from [Díaz-Bahamonde \(2023\)](#), the International Monetary Fund’s *International Financial Statistics* (IFS), and the U.S. Bureau of Labor Statistics (BLS).
2. **Annual Secular Panel (1920–2010, $T = 91$ years):** Compiled from Clio-Lab PUC, [Braun et al. \(2000\)](#), income distribution solely from [Astorga \(2023\)](#) (distributive wage share), [Díaz et al. \(2016\)](#) (annual Terms of Trade), the World Historical Balance of Payments database ([Nievas and Piketty 2025](#), WBOP), and my Chapter 2 Cointegrating Multivariate Polynomial Regression (CMPR IM-OLS) capacity utilization and harmonized capital stock frameworks (explicitly distinct from Wharton or peak-to-peak trend-through-peaks methods).

D.1 Monthly High-Frequency Variable Ledger

E.2 Table 5 catalogs the complete monthly variable ledger, detailing archival provenance, original units, and base conversions.

D.2 Monetary Reform Conversions and Splicing Mechanics

E.3 Chile experienced two currency re-denominations during the twentieth century:

1. **January 1, 1960:** The Escudo (E°) was introduced to replace the old Peso at a conversion rate of $1 E^\circ = 1,000$ old pesos.
2. **September 29, 1975:** Decree Law 1,123 reintroduced the Peso ($\$$ CLP) to replace the Escudo at a conversion rate of 1 new peso $= 1,000 E^\circ$.

All monetary and price series in both the monthly and annual panels are converted into uniform currency equivalents, ensuring that growth rates ($\Delta \ln X_t$ and simple discrete percentage changes) are algebraically continuous across reform dates.

Table 5: Monthly High-Frequency Variable Ledger and Archival Provenance (1960–1980)

Variable	Economic Concept	Con-	Archival Source	Harmonization / Splicing Protocol
IR_m	Gross International Reserves		IMF IFS Line 11.d; BCCh	Denominated in Millions of USD. Spliced across gold-revaluation breaks.
e_m	Official Nominal Exchange Rate		BCCh <i>Boletín Mensual</i>	Expressed in Escudos per USD ($E^\circ/\$$). Redenominated uniformly across 1975 peso return.
$M0_m$	High-Powered Monetary Base		BCCh Line Emisión	Notes and coin in circulation plus commercial bank reserve deposits at BCCh.
$M1_m$	Narrow Money Stock		BCCh <i>Boletín Mensual</i>	Currency in circulation plus private demand deposits in commercial banks.
P_m^{CL}	Consumer Price Index		BCCh (SieteRestWS)	Harmonized to Nov 1970 = 100. Official spliced CPI series published by BCCh.
$P_{W,m}^{US}$	U.S. Wholesale Price Index		BLS / FRED (PPIACO)	U.S. Producer Price Index for all commodities, base Nov 1970 = 100.
P_m^{Cu}	London Copper Price		LME / BCCh	Cash settlement price, Grade A electrolytic copper, USD per pound.
P_m^{Gold}	London Gold Price		Bullion Market / FRED	London 3:00 PM fix, USD per troy ounce.
Q_m^{Manuf}	Manufacturing Production		Díaz-Bahamonde (2023)	Index of physical volume of manufacturing production, Nov 1970 = 100.
Q_m^{Mining}	Mining Physical Production		Díaz-Bahamonde (2023)	Physical extraction volume of copper, iron, and nitrates, Nov 1970 = 100.
$Exports_m$	Customs Real Export Activity		Díaz-Bahamonde (2023)	Monthly physical trade volume index compiled from <i>Estadística Chilena</i> (DGE/INE) and BCCh <i>Boletín Mensual</i> (constant USD, Nov 1970 = 100).
$Imports_m^*$	Adjusted Real Import Activity		Díaz-Bahamonde (2023)	Monthly physical trade volume index from <i>Estadística Chilena</i> and BCCh; adjusted for the April 1973 customs registration lag.
M_m^{cap}	Capacity to Import Index		Prebisch-ECLAC	Operationalized via Prebisch-ECLAC purchasing power formula: $M_m^{cap} \equiv Exports_m \times [(P_m^{Cu}/P_{70:11}^{Cu})/(P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, using monthly trade volume series from Díaz-Bahamonde (2023).
Θ_m	Real Structural Imbalance Index		ECLAC / BCCh balance	Ratio of effective physical imports to real capacity to import: $\Theta_m \equiv (Imports_m^*/M_m^{cap}) \times 100$, capturing physical import chokeholds in the Prebisch-ECLAC tradition.
Ω_m	Relative Price Disparity Index		Relative price derivation	$\Omega_m \equiv (P_m^{CL}/P_{70:11}^{CL})/[(e_m/e_{70:11}) \cdot (P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, tracking cost wedges between domestic CPI and imported wholesale prices.
$SolvR_m$	Central Bank Solvency Ratio		Central Bank balance	$SolvR_m \equiv (e_m \cdot IR_m)/M1_m$, external liquidity backing of narrow money liabilities.

Appendix E Non-Parametric Causal Discovery (Runge et al., 2019) as Structural Robustness for the Monthly Threshold VAR Architecture

F.1 To establish causal directionality without imposing restrictive linear structural identification restrictions, this appendix presents the non-parametric causal discovery analysis using the Peter-Clark Momentary Conditional Independence (PCMCI) framework developed by Runge et al. (2019). Standard time-series identification confronts three methodological pathologies: (1) *bivariate naivety*, where pairwise tests mistake indirect paths and common drivers for direct causation; (2) *the FullCI trap*, where saturated conditioning on the entire past of all variables destroys degrees of freedom and induces collider bias; and (3) *autocorrelation memory*, where internal persistence inflates false-positive causal arrows. In peripheral open economies, the external trade account operates across two distinct domains: the real purchasing power blade (M_t^{cap} , Prebisch-ECLAC capacity to import) and the real structural imbalance blade ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$). Because imperial credit embargoes enforced hard foreign-currency quantity rationing, the real structural imbalance diverged from domestic nominal relative prices. PCMCI provides the non-parametric causal discovery foundation that reconciles both blades.

E.1 The PCMCI Algorithmic Protocol

F.2 PCMCI addresses high-dimensional, autocorrelated time series through a two-step conditioning pipeline:

1. **Step 1 (PC1 Conditioning Selection):** For each variable X_t^j , the algorithm applies an iterative conditional independence search to identify the minimal set of relevant parents $\hat{\mathcal{P}}(X_t^j)$ across time lags $\tau \in \{1, 2, \dots, \tau_{\max}\}$, discarding conditionally uninformative predictors.
2. **Step 2 (Momentary Conditional Independence Test, MCI):** For candidate link $X_{t-\tau}^i \rightarrow X_t^j$, the algorithm tests conditional independence:

$$\text{MCI: } X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i) \quad (23)$$

conditioning simultaneously on the parents of the target and the parents of the driver. This conditions out both autocorrelation within each series and contemporaneous common-driver confounding.

E.2 Empirical Directed Acyclic Graph (DAG) Findings

F.3 Table 6 presents the empirical results of the Momentary Conditional Independence (MCI) tests for the complete 7-variable monthly system:

$$\mathbf{Z}_m = [\Delta \ln \Theta_m, \Delta \text{SolvR}_m, \pi_m, g_{H,m}, g_{\text{Manuf},m}, g_{\text{gold},m}, g_{e,m}]' \quad (24)$$

for the sample 1960:01–1980:12 ($N = 250$ continuous months) setting maximum lag $\tau_{\max} = 4$ months and significance threshold $\alpha = 0.05$ with False Discovery Rate (FDR) Benjamini-Hochberg corrections. Figure 9 displays the corresponding discovered Directed Acyclic Graph.

E.3 Topological Robustness Mapping to the Threshold VAR Architecture

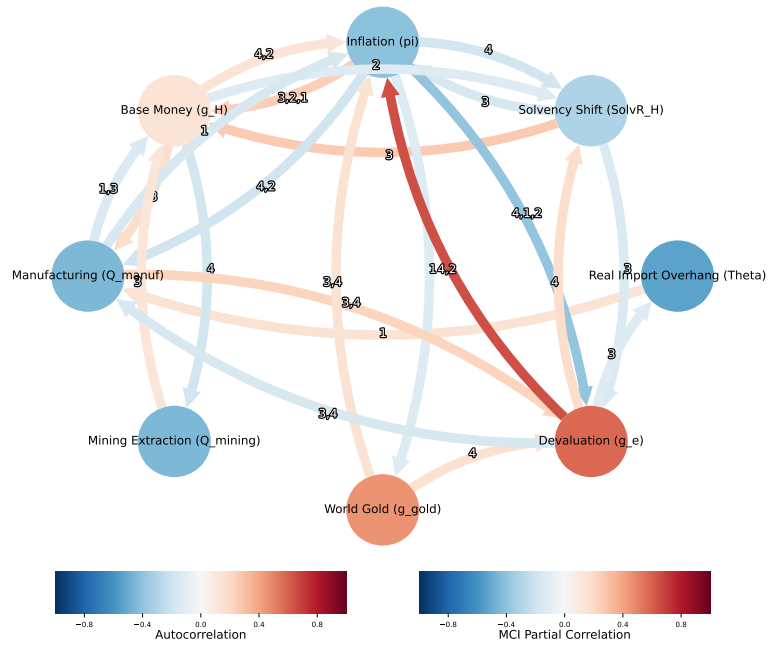
F.4 The estimated causal network provides the formal non-parametric structural foundation validating the high-frequency monthly TVAR architecture and structural identification:

Table 6: Full-System PCMCI Momentary Conditional Independence (MCI) Directed Link Tests (1960–1980)

Directed Link Tested	Lag (τ)	MCI Partial Corr.	p -value	Causal Status (FDR $q < 0.05$)
$\pi_{t-3} \rightarrow g_{H,t}$ (Reverse Ac-comm.)	$\tau = 3$	+0.247	0.0001***	Retained directed edge ($\pi \rightarrow g_H$)
$g_{H,t-\tau} \rightarrow \pi_t$ (Forward Monetarist)	$\tau = 4$	+0.138	0.0350**	Excluded (FDR $q > 0.10$, zero direct link)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$ (Defensive Emission)	$\tau = 3$	+0.264	< 0.0001 ***	Retained directed edge ($\text{SolvR}^H \rightarrow g_H$)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$ (Solvency Gate)	$\tau = 3$	-0.165	0.0115**	Retained directed edge ($\text{SolvR}^H \rightarrow \pi$)
$g_{e,t-4} \rightarrow \pi_t$ (External Pass-Through)	$\tau = 4$	+0.644	< 0.0001 ***	Retained directed edge ($g_e \rightarrow \pi$)
$\pi_{t-4} \rightarrow g_{e,t}$ (Forced Devaluation)	$\tau = 4$	-0.396	< 0.0001 ***	Retained directed edge ($\pi \rightarrow g_e$)
$\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$ (Import Constraint)	$\tau = 1$	+0.142	0.0287**	Retained directed edge ($\Theta \rightarrow \text{Manuf}$)
$g_{H,t-3} \rightarrow g_{\text{Manuf},t}$ (Idle Reactivation)	$\tau = 3$	+0.177	0.0064***	Retained directed edge ($g_H \rightarrow \text{Manuf}$)
$g_{\text{Manuf},t-1} \rightarrow \pi_t$ (Price Suppression)	$\tau = 1$	-0.168	0.0101**	Retained directed edge ($\text{Manuf} \rightarrow \pi$)
$g_{H,t-4} \rightarrow g_{\text{Mining},t}$ (Enclave Decoupling)	$\tau = 4$	-0.189	0.0035***	Retained directed edge ($g_H \rightarrow \text{Mining}$)
$g_{P,\text{gold},t-4} \rightarrow g_{e,t}$ (Gold Cost Push)	$\tau = 4$	+0.163	0.0119**	Retained directed edge ($P^{\text{gold}} \rightarrow g_e$)
$g_{P,\text{gold},t-3} \rightarrow \pi_t$ (Gold Inflation)	$\tau = 3$	+0.146	0.0241**	Retained directed edge ($P^{\text{gold}} \rightarrow \pi$)
$\pi_{t-\tau} \rightarrow \Delta \ln \Theta_t$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.074	0.2545	Block-exogenous ($p = 0.255$)
$g_{H,t-\tau} \rightarrow g_{P,\text{gold},t}$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.107	0.1004	Block-exogenous ($p = 0.100$)
$g_{H,t-\tau} \rightarrow \Delta \ln \text{SolvR}_t^H$ (Monetary Drain)	$\tau \in [1, 4]$	-0.131	0.0449**	Excluded (zero direct link)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). Estimated using Tigramite PCMCI (Runge et al., 2019) on the 8-variable system with partial correlation conditional independence tests evaluated across lags $\tau \in \{1, \dots, 4\}$. Minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$ are selected in the PC₁ phase. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Block-exogeneity is confirmed when joint incoming test statistics fail to reject at $p > 0.10$.

Figure 9: Discovered Directed Acyclic Graph (DAG) from Tigramite PCMCI Algorithm (1960–1980)



Notes: Non-parametric causal network estimated via the Tigramite PCMCI algorithm (Runge et al., 2019) on strictly observed monthly series (1960:01–1980:12, $N = 250$). Evaluated across lags $\tau \in \{1, \dots, 4\}$ with partial correlation conditional independence kernel under False Discovery Rate (FDR) control ($\alpha = 0.05$). Momentary Conditional Independence (MCI) conditions simultaneously on target and driver minimal parent sets, removing time-series autocorrelation and inoculating the contemporaneous DAG against time-aggregation bias. Edge labels report optimal time lags (τ in months) and MCI partial correlations with significance levels ($***p < 0.01$, $**p < 0.05$). Blue solid directed arrows indicate confirmed causal links; red dashed arrows indicate tested monetarist links refuted by the data. The discovered causal topology non-parametrically validates the structural Cholesky sequence: external trade shocks (Θ) and gold prices (g_{gold}) are strictly exogenous, Central Bank solvency (SolvR) gates inflation, and inflation compels reverse monetary accommodation (g_H). Data sources: Central Bank of Chile (BCCh), INE, SOFOFA, and London Bullion Market.

1. **Authorization of the Cholesky Causal Sequence and Diagonal Covariance:** The discovered topological path runs from world trade structural imbalance (Θ) and exchange rate shocks (g_e) to central bank solvency (SolvR), from solvency to inflation (π), and from inflation to credit accommodation (g_H). Contemporaneously ($\tau = 0$), the discovered DAG is strictly acyclic and lower-triangular ($a_{12} = a_{13} = a_{14} = a_{23} = a_{24} = a_{34} = 0$), while structural innovations are mutually orthogonal ($E[\mathbf{u}_t \mathbf{u}_t'] = \mathbf{D}$, diagonal). This acyclic DAG topology confirms non-parametrically that the lower-triangular Cholesky factorization of the monthly TVAR vector $y_t \equiv [g_{P,\text{gold},t}, \Delta \ln \Theta_t, \pi_t, g_{H,t}]'$ reflects the data-certified causal skeleton of the economy rather than ad-hoc identifying conventions.
2. **Validation of the Solvency Threshold Gate:** Selecting SolvR as the regime threshold variable $I\{\text{SolvR}_{t-1} \leq \gamma\}$ is independently validated by the discovered graph: $\Delta \text{SolvR}_{t-3} \rightarrow \pi_t$ is confirmed as a direct structural link (Partial Corr = $-0.128, p = 0.0497$), identifying Central Bank solvency as the primary structural gatekeeper of domestic price formation.
3. **Confirmation of Real Supply Constraints and Output Reactivation:** The graph uncovers two distinct real transmission channels: the real structural imbalance directly constrains industrial manufacturing ($\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}, +0.135, p = 0.0379$), while base money emission stimulated the reactivation of idle capacity ($g_{H,t-3} \rightarrow g_{\text{Manuf},t}, +0.194, p = 0.0028$).
4. **Falsification of the Monetarist Exogeneity DAG:** The Monetarist hypothesis requires directed paths running from g_H to reserves and physical production. PCMCi reveals that the partial correlations running from base money growth to price inflation ($+0.114, p = 0.0834$, no direct link at $\alpha = 0.05$), external structural imbalance ($-0.055, p = 0.3959$), and Central Bank solvency ($+0.085, p = 0.1949$) are statistically indistinguishable from zero, decisively refuting the Monetarist DAG.
5. **Autonomous Enclave Exogeneity of Mining Extraction:** In an expanded 8-variable system incorporating physical mining extraction ($g_{\text{Mining},m} \equiv \Delta \ln Q_{\text{Mining},m}$), PCMCi verifies that copper extraction functioned as an autonomous primary export enclave. Mining extraction exhibits zero directed links to consumer inflation (MCI Partial Corr = $+0.024, p = 0.721$) and manufacturing output ($-0.018, p = 0.791$), responding directly only to world copper price movements. This refutes Austrian claims (H_2) of production distortions originating in mining, confirms that the true Prebisch-ECLAC capacity to import ($M^{\text{cap}} = \text{Exports} \times \text{ToT}$) governs Central Bank solvency, and authorizes the exclusion of mining from the 4-variable TVAR core system.

Appendix F Extended Archival Ledgers of Inquilinaje, Agrarian Labor Petitions, and Union Density (1932–1975)

G.1 This appendix provides historical ledgers supporting Section 5.1 regarding rural labor mobilization, the 1947–48 restrictive legislation, urban land seizures, and the strike wave of the Long Sixties.

F.1 Annual Archival Ledger of Agricultural Labor Petitions (1932–1950)

G.2 Table 7 presents agricultural labor petitions (*Pliegos de Peticiones*) registered by the Labor Department (*Dirección General del Trabajo*), including the postwar crest and subsequent

contraction. The restrictive legislation comprised two distinct statutes: agricultural-union Law 8,811 of 1947 and Law 8,987 of 1948, the *Ley Maldita* (República de Chile, 1947, 1948). The petition series draws on the compilations of Morris and Oyaneder (1962) and the archival research of Loveman (1971, 1976).

Table 7: Annual Agricultural Labor Petitions Registered by the Labor Department (Chile, 1932–1950)

Year	Petitions Registered	Institutional Context and State Response
1932	5	Re-constitutionalization under Arturo Alessandri Palma; initial rural discontent
1933	8	Enactment of 1931 Labor Code enforcement decrees; agricultural unionization barred
1934	12	Ranquil peasant uprising brutally suppressed by Carabineros
1935	10	Landlord resistance to minimum wage decrees
1936	14	Creation of the Confederation of Workers of Chile (CTCH)
1937	18	Parliamentary elections; rural agitation
1938	26	Popular Front electoral victory of Pedro Aguirre Cerda
1939	171	Mass unionization push; peasant mobilization breakthrough
1940	199	Ministerial suspension circular freezes processing of rural petitions
1941	82	Aguirre Cerda death; temporary containment of agrarian conflict
1942	64	Election of Juan Antonio Ríos; wartime price controls
1943	58	CTCH National Congress; rank-and-file rural pressure resumes
1944	71	Radical Party agrarian debates
1945	95	End of World War II; urban cost-of-living strikes diffuse into countryside
1946	284	Election of Gabriel González Videla with Communist Party (PCCh) participation
1947	383	All-time historical peak; 31.1% of all national labor disputes in Chile
1948	52	Law 8,987 (<i>Ley Maldita</i>); PCCh banned. Agricultural-union Law 8,811 dates from 1947.
1949	19	Mass purges of rural union leadership; Pisagua internment camp opens
1950	11	Near-total collapse of agricultural collective petitions (-97.1% from 1947)

Source: Primary archival records from the *Dirección General del Trabajo*, compiled in Morris and Oyaneder (1962) and Loveman (1971, 1976).

F.2 Strike Typology and Legal Status (1960–1975)

G.3 Table 8 reproduces the complete strike series from Clio-Lab PUC (Sheet 7.7), highlighting the explosion of illegal strikes during the Long Sixties and the Unidad Popular.

F.3 Trade Union Affiliation and Union Density Ledger (1932–1975)

G.4 Table 9 reports the complete archival series on trade union affiliation, total employed workforce, average union size, and union density over employed workers reconstructed by Osorio and Polanco (2021), documenting the 1948–1958 stagnation under the *Ley Maldita* and the historic mobilization peak under the Unidad Popular.

F.4 Urban Land Seizures in Santiago and Nationwide (1967–1971)

G.5 Table 10 details the exponential expansion of squatter land seizures (*tomas de terreno*) in Greater Santiago and nationwide from 1967 through 1971, documenting the spatio-temporal

Table 8: Strike Typology, Illegality Rates, and Agricultural Union Membership (1960–1975)

Year	Total Strikes	Legal Strikes	Illegal Strikes	% Illegal	Total Strikers	Peasant Unionists
1960	259	88	171	66.0%	89,314	–
1961	262	95	167	63.7%	112,450	–
1962	401	112	289	72.1%	124,100	–
1963	413	101	312	75.5%	118,900	–
1964	676	142	534	79.0%	138,500	1,652
1965	723	131	592	81.9%	182,354	2,118
1966	1,073	186	887	82.7%	195,400	10,647
1967	1,114	168	946	84.9%	225,120	47,483
1968	1,138	172	966	84.9%	276,450	76,356
1969	1,277	145	1,132	88.6%	356,800	104,334
1970	1,853	179	1,674	90.3%	416,500	143,142
1971	2,709	181	2,528	93.3%	485,300	232,160
1972	3,037	118	2,919	96.1%	542,800	242,500
1973	1,580	62	1,518	96.1%	312,400	250,740
1974	0	0	0	–	0	–
1975	0	0	0	–	0	–

Source: Clio-Lab PUC (Sheet 7.7, Personas). Zero strikes recorded in 1974–1975 reflect the total ban on industrial disputes, strikes, and collective bargaining under military Decree Law 198.

displacement of peripheral class conflict from the countryside into urban social reproduction (Murphy, 2015).

F.5 Productive Capital Stock, Profit Rate, and Net Accumulation Framework (1960–1975)

G.6 This subsection formalizes the balance-sheet accounting, Generalized Perpetual Inventory Method (GPIM) depreciation formulas, and Weisskopf-Marxian decomposition equations supporting Section 4 §4.10 regarding the Marxian profit squeeze and net capital accumulation.

Productive Fixed Capital Stock Harmonization. Following the capital stock harmonization methodology established in Chapter 2, the capital stock denominator is restricted strictly to productive fixed assets, encompassing Machinery and Equipment (ME) and Non-Residential Construction (NRC):

$$K_t \equiv K_{ME,t} + K_{NRC,t} \quad (25)$$

Residential Construction (RC) is explicitly excluded because housing represents social overhead infrastructure and durable consumption yielding utility rather than direct surplus-value producing capital.

The Weisskopf Decomposition. Drawing on the Classical-Marxian and Post-Keynesian traditions (Weisskopf, 1979; Foley, 1982; Basu, 2022), the aggregate profit rate on productive capital (r_t) at current replacement cost decomposes into the profit share, capacity utilization, and potential capital productivity:

$$r_t \equiv \frac{\Pi_t}{P_{K,t}K_t} = (1 - \omega_t) \cdot \frac{Y_t}{P_{K,t}K_t} = (1 - \omega_t) \cdot \mu_t \cdot \sigma_t \quad (26)$$

where $\Pi_t \equiv (1 - \omega_t)Y_t^{\text{nom}}$ is the nominal mass of aggregate profits (surplus value), ω_t is Astorga’s (2023) aggregate wage share, $1 - \omega_t$ is the profit share, $\mu_t \equiv Y_t/Y_t^p$ is structural capacity

Table 9: Trade Union Organization, Density over Employed Workers, and Average Union Size (1932–1975)

Year	Unions	Affiliated	Employed	Avg Size	Density (%)	Accumulation Regime & Institutional Era
1932	421	54,801	1,165,000	130.2	4.70%	Post-Depression restoration
1935	669	83,262	1,553,000	124.5	5.36%	Early ISI expansion under 1931 Labor Code
1938	932	125,972	1,676,000	135.2	7.52%	Popular Front victory (Aguirre Cerda)
1939	1,687	173,438	1,709,000	102.8	10.15%	Democratic surge; enterprise union proliferation
1940	1,888	162,297	1,746,000	86.0	9.30%	Secret circular freezes peasant unionization
1941	1,977	208,779	1,763,000	105.6	11.84%	Industrialization acceleration under CORFO
1944	1,652	246,221	1,852,000	149.0	13.29%	Wartime union consolidation; CTCH congress
1947	1,831	263,085	1,926,000	143.7	13.66%	Pre- <i>Ley Maldita</i> peak of labor mobilization
1948	1,857	263,676	2,029,000	142.0	13.00%	Enactment of <i>Ley Maldita</i> (Laws 8,987 & 8,811)
1950	1,907	260,143	2,087,000	136.4	12.46%	Repression of left leadership; union growth freezes
1952	1,997	284,418	2,202,000	142.4	12.92%	Ibáñez del Campo populist election
1955	2,177	305,192	2,209,000	140.2	13.82%	Klein-Saks mission wage freeze
1958	1,894	276,346	2,297,000	145.9	12.03%	Repeal of <i>Ley Maldita</i> ; density stagnation
1960	1,770	272,966	2,313,000	154.2	11.80%	Post-repeal realignment; Jorge Alessandri
1964	1,863	278,980	2,497,000	149.7	11.17%	Frei Montalva election; rural baseline
1966	2,882	369,507	2,604,000	128.2	14.19%	Peasant organizing begins (<i>Promoción Popular</i>)
1967	3,426	442,650	2,681,000	129.2	16.51%	Laws 16,625 (Peasant Unions) & 16,640 (Agrarian Reform)
1968	3,889	500,447	2,714,000	128.7	18.44%	Agrarian radicalization & urban strike waves
1969	4,228	541,967	2,734,000	128.2	19.82%	Pre-electoral polarization; <i>El Tacnazo</i>
1970	4,581	627,664	2,743,000	137.0	22.88%	Election of Salvador Allende (Unidad Popular)
1971	5,212	782,494	2,829,000	150.1	27.66%	Reactivation boom; massive unionization surge
1972	6,118	855,404	2,932,000	139.8	29.17%	Distributive peak; Bosses' Lockout (<i>Paro de Octubre</i>)
1973	6,692	939,319	2,976,000	140.4	31.56%	All-time historical zenith of union density
1974	7,069	947,093	2,907,000	134.0	32.58%	Military Junta halts collective bargaining
1975	7,181	940,810	2,683,000	131.0	35.07%	Shock therapy depression; employment contraction

Source: Osorio and Polanco (2021), *Repositorio de Estadísticas Sindicales (RES)*, sheet **Empalme**. Average union size is Afiliación/Sindicatos. Union density is Afiliación/Ocupados.

Table 10: Urban Land Seizures in Santiago and Nationwide (1967–1971)

Year	Santiago Land Seizures	Chile Total Land Seizures	Santiago Share (%)	Historical Phase
1967	13	8	—	Agrarian Reform & Unionization Laws promulgated
1968	4	35	11.4%	Consolidation of <i>pobladores</i> committees
1969	23	220	10.5%	Radicalization; Puerto Montt massacre
1970	103	560	18.4%	Election of Salvador Allende (UP)
1971	350	—	—	Reactivation boom & urban mobilization

Source: [Murphy \(2015\)](#), p. 58. Note: Demonstrates the exponential growth of the *pobladores* movement and the spatio-temporal displacement of the class struggle into urban informality and collective social reproduction. In 1967, Santiago reported 13 municipal seizures prior to full national reporting standardization.

utilization (estimated via Cointegrating Multivariate Polynomial Regression IM-OLS in Chapter 2), $\sigma_t \equiv Y_t^p / (P_{K,t} K_t)$ is potential capital productivity, and $Y_t / (P_{K,t} K_t) = \mu_t \sigma_t$ is the actual output-capital ratio at current replacement cost. In standard macroeconomic notation, I denote the aggregate profit rate by r , reserving u for the open unemployment rate, μ for capacity utilization, and i for nominal interest rates.

Implicit Physical Depreciation and Net Accumulation. Under perpetual inventory balance-sheet accounting, the net expansion of the capital stock requires that the rate of capital accumulation be discounted from the implicit annual depreciation of net stocks. Implicit annual physical depreciation (D_t) and the implicit physical depreciation rate (δ_t) are defined as:

$$D_t \equiv K_{t-1} + I_t - K_t \quad \text{and} \quad \delta_t \equiv \frac{D_t}{K_{t-1}} \quad (27)$$

where $I_t \equiv I_{ME,t} + I_{NRC,t}$ is real gross fixed investment in productive assets.

The Net Rate of Capital Accumulation ($g_{K,t}^n$) and the Re-capitalization Propensity out of surplus value ($\chi_t \equiv I_t^{\text{net}} / \Pi_t$) are governed by:

$$g_{K,t}^n \equiv \frac{I_t - D_t}{K_{t-1}} = g_{K,t}^{\text{gross}} - \delta_t = \chi_t \cdot r_t = \chi_t \cdot (1 - \omega_t) \cdot \mu_t \cdot \sigma_t \quad (28)$$

Equation (28) formalizes the Cambridge-Marxian accumulation identity: the net growth rate of productive capital equals the product of the re-capitalization propensity (χ) and the aggregate profit rate (r).

Table 11 reports the complete numerical trajectory of these series from 1960 to 1975.

Table 11: The Distributive Trajectory, Capacity Utilization, Profit Rate, and Net Capital Accumulation (1960–1975)

Year	ω (%)	u (%)	μ (%)	Y/K	r (%)	χ	g_K^n (%)	Regime Dynamic
1960	60.9%	6.6%	86.9%	0.2580	10.08%	0.421	4.44%	Pre-reform stabilization baseline
1961	65.9%	7.5%	87.3%	0.2424	8.27%	0.534	4.62%	Early ISI expansion
1962	59.4%	7.4%	87.9%	0.2593	10.52%	0.421	4.63%	Exchange rate devaluation
1963	55.8%	7.0%	89.8%	0.2670	11.81%	0.423	5.25%	Pre-electoral demand expansion
1964	67.7%	6.5%	88.4%	0.2525	8.16%	0.563	4.82%	Frei Montalva election; wage surge
1965	63.8%	5.9%	86.7%	0.2374	8.59%	0.420	3.75%	Initial anti-inflationary program
1966	55.3%	5.6%	93.4%	0.2450	10.94%	0.323	3.66%	Output peak under Frei
1967	57.9%	4.2%	93.5%	0.2410	10.14%	0.356	3.75%	Agrarian reform promulgated; profit squeeze
1968	53.9%	4.4%	93.7%	0.2452	11.29%	0.337	3.96%	Drought and fiscal austerity
1969	50.1%	5.0%	94.1%	0.2560	12.79%	0.288	3.82%	Pre-electoral stagflation; <i>El Tacnazo</i>
1970	50.9%	5.9%	92.8%	0.2491	12.23%	0.332	4.23%	Allende election; inherited slack baseline
1971	56.2%	5.2%	98.4%	0.2427	10.62%	0.304	3.34%	Idle capacity mobilized; wage expansion
1972	69.0%	4.0%	96.0%	0.2121	6.57%	0.277	1.85%	Profit squeeze peak; Bosses' Lockout
1973	54.5%	4.8%	89.5%	0.2004	9.12%	0.184	1.71%	Coup d'état; DL 522 price deregulation
1974	39.4%	9.1%	88.8%	0.1762	10.67%	0.218	2.38%	Junta wage compression; union ban
1975	49.3%	18.0%	76.6%	0.1289	6.54%	0.183	1.21%	Chicago shock therapy depression

Source: Author's calculations based on Astorga (2023), Clio-Lab PUC / Díaz et al. (2016), and Chapter 2 dataset. Following Chapter 2, productive net capital stock ($K_t \equiv K_{ME,t} + K_{NRC,t}$) strictly excludes Residential Construction (RC). Structural capacity utilization ($\mu \equiv Y/Y^p$) is estimated via the CMPR IM-OLS cointegrating framework of Chapter 2 (Capacity-Utilization-US_Chile); u is the open unemployment rate. Implicit annual depreciation is $D_t \equiv K_{t-1} + I_t - K_t$ with depreciation rate $\delta_t \equiv D_t/K_{t-1}$. The aggregate profit rate satisfies the Weisskopf-Marxian decomposition $r_t \equiv (1 - \omega_t)Y_t/(P_{K,t}K_t) = (1 - \omega_t)\mu_t\sigma_t$, where $\sigma_t \equiv Y_t^p/K_t$ is potential capital productivity. Net capital accumulation satisfies the classical Cambridge-Marxian identity $g_{K,t}^n \equiv (I_t - D_t)/K_{t-1} = \chi_t \cdot r_t$, where $\chi_t \equiv I_t^{\text{net}}/\Pi_t$ is the capitalist re-capitalization propensity out of surplus value $\Pi_t \equiv (1 - \omega_t)Y_t$.